

## Research Paper

## Subspace projection regularization for large-scale Bayesian linear inverse problems

Haibo Li *School of Mathematics and Statistics, The University of Melbourne, Melbourne, VIC, 3010, Australia*

## ARTICLE INFO

## 2000 MSC:

62F15

65J22

65F22

65F10

## Keywords:

Bayesian inverse problem

Gaussian prior

Subspace projection regularization

Generalized Golub-Kahan bidiagonalization

Semi-convergence

Early stopping

## ABSTRACT

The Bayesian statistical framework provides a systematic approach to enhance the regularization model by incorporating prior information about the desired solution. For the Bayesian linear inverse problems with Gaussian noise and Gaussian prior, we propose an iterative regularization algorithm within the subspace projection regularization (SPR) framework. By treating the forward model matrix as a linear operator between two proper finite dimensional Hilbert spaces, we first introduce an iterative process that can generate a series of valid solution subspaces. The SPR method then projects the original problem onto these solution subspaces to get a series of low dimensional linear least squares problems, where an efficient procedure is developed to update the solutions of them to approximate the desired solution of the original problem. With the newly designed early stopping rules, this iterative algorithm can obtain a regularized solution with satisfied accuracy. We establish several theoretical results to reveal the regularization properties of the algorithm. Both small-scale and large-scale inverse problems are used to test the proposed algorithm and demonstrate its robustness and efficiency. The most computationally intensive operations in the proposed algorithm only involve matrix-vector products, making it highly efficient for large-scale problems.

## 1. Introduction

Inverse problems arise frequently in various scientific and engineering fields such as medical imaging, computed tomography, geoscience, data assimilation and so on [1–5]. In these problems, we aim to reconstruct unknown parameters or functions from indirect measurements that often include noise. In this paper, we consider linear inverse problems of the discrete form

$$\mathbf{b} = \mathbf{A}\mathbf{x} + \boldsymbol{\epsilon}, \boldsymbol{\epsilon} \sim \mathcal{N}(\mathbf{0}, \mathbf{M}), \quad (1.1)$$

where  $\mathbf{x} \in \mathbb{R}^n$  is the underlying quantity we wish to reconstruct,  $\mathbf{A} \in \mathbb{R}^{m \times n}$  is the discretized forward model matrix,  $\mathbf{b} \in \mathbb{R}^m$  is the vector of observation, and  $\boldsymbol{\epsilon}$  is the noise that follows a zero mean Gaussian distribution with positive definite covariance matrix  $\mathbf{M}$ . In the sense of Hadamard [6], this problem is usually ill-posed, which means that there may be multiple solutions that fit the observation equally well, or the solution is very sensitive respect to observation perturbation. The ill-posedness of inverse problems is a big challenge that prohibits us from computing an accurate solution by a direct inversion method.

Regularization is a very common technique to overcome ill-posedness, where the core idea is to use prior knowledge about the underlying solution to constrain the set of possible solutions and improve their stability and uniqueness properties. Here we deal with the inverse problem (1.1) and its regularization from a Bayesian perspective [1,7], where the unknown and observation are modeled as random variables, and prior knowledge and assumptions about the unknown are combined with the observation to obtain a posterior

E-mail address: [haibo.li@unimelb.edu.au](mailto:haibo.li@unimelb.edu.au)

<https://doi.org/10.1016/j.apnum.2025.10.016>

Received 19 August 2024; Received in revised form 6 September 2025; Accepted 24 October 2025

Available online 27 October 2025

0168-9274/© 2025 Published by Elsevier B.V. on behalf of IMACS.

probability distribution of  $\mathbf{x}$ . By (1.1), the observation vector  $\mathbf{b}$  has a conditional probability density function (pdf)

$$p(\mathbf{b}|\mathbf{x}) \propto \exp\left(-\frac{1}{2}\|\mathbf{Ax} - \mathbf{b}\|_{\mathbf{M}^{-1}}^2\right).$$

Maximizing  $p(\mathbf{b}|\mathbf{x})$  with respect to  $\mathbf{x}$  is known as the maximum likelihood estimate (MLE), where  $\|\mathbf{x}\|_{\mathbf{B}} := (\mathbf{x}^\top \mathbf{B} \mathbf{x})^{1/2}$  is the  $\mathbf{B}$ -norm of  $\mathbf{x}$  for a positive definite matrix  $\mathbf{B}$ . To regularize MLE, we assume a Gaussian prior about the desired solution with the form  $\mathbf{x} \sim \mathcal{N}(\mathbf{0}, \lambda^{-1} \mathbf{N})$ , where  $\mathbf{N}$  is a positive definite covariance matrix. Using the Bayes' formula that estimates posterior pdf of  $\mathbf{x}$  after observation  $\mathbf{b}$ , we have

$$p(\mathbf{x}|\mathbf{b}, \lambda) \propto p(\mathbf{x}|\lambda)p(\mathbf{b}|\mathbf{x}) \propto \exp\left(-\frac{1}{2}\|\mathbf{Ax} - \mathbf{b}\|_{\mathbf{M}^{-1}}^2 - \frac{\lambda}{2}\|\mathbf{x}\|_{\mathbf{N}^{-1}}^2\right).$$

The maximum a posterior estimate (MAP) of  $\mathbf{x}$  finds a regularized solution to (1.1) by maximizing the posterior pdf of  $\mathbf{x}$ , which leads to

$$\min_{\mathbf{x} \in \mathbb{R}^n} \{\|\mathbf{Ax} - \mathbf{b}\|_{\mathbf{M}^{-1}}^2 + \lambda\|\mathbf{x}\|_{\mathbf{N}^{-1}}^2\}. \quad (1.2)$$

The form of (1.2) is the same as the Tikhonov regularization [8], where the regularization term  $\|\mathbf{x}\|_{\mathbf{N}^{-1}}^2$  enforces extra structure on the solution that comes from the prior distribution of  $\mathbf{x}$ . The regularization parameter  $\lambda$  controls the trade-off between the data-fit term and regularization term, which should be determined in advance.

The Bayesian statistical framework offers a systematic approach to enhance the regularization model by incorporating prior information about the solution. However, developing efficient algorithms for computing those informative solutions is very challenging. Firstly, the quality of the reconstructed solution heavily relies on the accuracy of the regularization parameter. Although various methods have been proposed, such as the L-curve (LC) method [9], generalized cross-validation (GCV) [10] and discrepancy principle (DP) [11], but each has its limitations and assumptions, and these methods require solving (1.2) for many different values of  $\lambda$ . Secondly, to solve (1.2), the Cholesky factorizations  $\mathbf{M}^{-1} = \mathbf{L}_M^\top \mathbf{L}_M$  and  $\mathbf{N}^{-1} = \mathbf{L}_N^\top \mathbf{L}_N$  are required to transform (1.2) to the standard 2-norm form

$$\min_{\mathbf{x} \in \mathbb{R}^n} \{\|\mathbf{L}_M(\mathbf{Ax} - \mathbf{b})\|_2^2 + \lambda\|\mathbf{L}_N \mathbf{x}\|_2^2\} \quad (1.3)$$

if an iterative solver for the above least squares problem is exploited; or  $\mathbf{M}^{-1}$  and  $\mathbf{N}^{-1}$  must be computed if an iterative solver for the normal equation of (1.2) is exploited. For both the cases, it is not feasible to work with the matrix inversion and factorization if its dimension is very large or the explicit form of the matrix is not available.

Up to now, there are many numerical methods that deal with large-scale Bayesian linear inverse problems; here we review iterative methods that are based upon Krylov subspaces [12]. The main idea behind these methods is to project the original linear system (1.1) onto a series of suitable low dimensional subspaces and solve these small projected problems step by step, and the iteration should be terminated early to avoid too many noisy components included in the solution [1,13]. In this approach, a key point is that these subspaces must be elaborately constructed by an iterative process that can incorporate information about the prior distribution of  $\mathbf{x}$ . To this end, the priorconditioning technique is proposed, where the statistically inspired left and right priorconditioners are used to whiten the noise and unknown, respectively [13–15]. However, it usually requires that  $\mathbf{M}^{-1}$  and its Cholesky factorization can be computed easily, and  $\mathbf{L}_N$  instead of  $\mathbf{N}^{-1}$  should be explicitly constructed as the right priorconditioner. In some scenarios of applications, the covariance matrix  $\mathbf{N}$  instead of  $\mathbf{L}_N$  is used to add prior information about  $\mathbf{x}$ , such as the Metérn class of covariance matrices [16–18]. This raises the need for new methods to construct suitable solution subspaces to incorporate prior information about  $\mathbf{x}$  while avoiding Cholesky factorizations and inversions of the prior covariance matrices.

Recently, a generalized hybrid iterative method is proposed for large-scale Bayesian linear inverse problems [19], which is based on an iterative process called the generalized Golub-Kahan bidiagonalization (gen-GKB) [20]. This method first transforms (1.2) to

$$\min_{\tilde{\mathbf{x}} \in \mathbb{R}^n} \{\|\mathbf{AN}\tilde{\mathbf{x}} - \mathbf{b}\|_{\mathbf{M}^{-1}}^2 + \lambda\|\tilde{\mathbf{x}}\|_{\mathbf{N}}^2\} \quad (1.4)$$

by the substitution  $\mathbf{x} = \mathbf{N}\tilde{\mathbf{x}}$  to avoid the inversion of  $\mathbf{N}$ . Then it uses gen-GKB to generate a series of  $N$ -orthonormal bases of  $k$ -dimensional Krylov subspaces for  $k = 1, 2, \dots$  and projects (1.4) onto these subspaces to get a series of  $k$ -dimensional Tikhonov regularization problems. To avoid choosing a good regularization parameter in advance, this method adapts a hybrid regularization approach, which determines a regularization parameter  $\lambda_k$  for the projected  $k$ -dimensional problem at each  $k$ -th step using a parameter choosing criterion such as GCV [21]. By determining  $\lambda_k$  at each iteration, this method computes a series of iterative solutions which are used to approximate the best solution to (1.4) with the optimal parameter  $\lambda_{opt}$ .

The gen-GKB based hybrid regularization method is reasonable in the sense that the iterative solution lies within the same subspace as the MAP estimator of the original Bayesian inverse problem (1.2), thereby the prior information of  $\mathbf{x}$  encoded by the Gaussian prior can be well incorporated into the iterative solution. Furthermore, the computations of  $\mathbf{N}^{-1}$  and factorizations of  $\mathbf{M}^{-1}$  and  $\mathbf{N}^{-1}$  are not needed, which makes it very practical for solving large-scale inverse problems. Although this method is effective for many types of inverse problems, sometimes the regularization parameter  $\lambda_k$  can not be well determined, resulting in that  $\lambda_k$  does not converge to  $\lambda_{opt}$  or converges very slowly, and the corresponding iterative solution have a poor accuracy; this is a common potential flaw for hybrid methods [22,23]. Besides, this method needs to transform (1.2) to (1.4) first to apply gen-GKB, and there is not a direct motivation provided to explain why gen-GKB should be used to generate valid solution subspaces for (1.2).

In order to avoid choosing regularization parameter in advance for (1.2) or step by step for the generalized hybrid method, we consider the iterative subspace projection regularization (SPR) method [24, §3.3]. For the Bayesian linear inverse problem (1.2), the

core idea is to seek a series of  $\mathbf{x}_k$  as the solution to

$$\min_{\mathbf{x} \in \mathcal{X}_k} \|\mathbf{x}\|_{N^{-1}}, \quad \mathcal{X}_k = \{\mathbf{x} : \min_{\mathbf{z} \in S_k} \|\mathbf{A}\mathbf{x} - \mathbf{b}\|_{M^{-1}}\},$$

where  $S_k$  is the subspace of  $\mathbb{R}^n$  of dimension  $k = 1, 2, \dots$ , and the iteration proceeds until an early stopping rule is satisfied to overcome too many noisy components are contained. We call  $S_k$  the *solution subspace*, which should be constructed elaborately such that the information about the Gaussian noise  $\epsilon$  and desired property of  $\mathbf{x}$  encoded by the Gaussian prior can be incorporated into it. In this paper, we propose an iterative regularization algorithm within the SPR framework based on an iterative process that is equivalent to gen-GKB. This new algorithm exhibits good performance for indirectly solving the Bayesian linear inverse problem (1.2) and is very efficient for large-scale problems.

The main contributions of this paper are listed as follows:

- By treating the matrix  $\mathbf{A}$  as a bounded linear operator between the two Hilbert spaces  $(\mathbb{R}^n, \langle \cdot, \cdot \rangle_{N^{-1}})$  and  $(\mathbb{R}^m, \langle \cdot, \cdot \rangle_{M^{-1}})$ , which are equipped with the  $N^{-1}$ - and  $M^{-1}$ -inner products, we introduce an iterative process that can generate a series of  $N^{-1}$ -orthonormal bases of  $k$ -dimensional subspaces of  $(\mathbb{R}^n, \langle \cdot, \cdot \rangle_{N^{-1}})$ . This process is mathematically equivalent to gen-GKB, thereby we use the same name for it. The use of the new inner products instead of the standard ones provides a very natural view for developing iterative regularization methods for Bayesian linear inverse problems.
- With the solution subspace spanned by the  $N^{-1}$ -orthonormal basis generated by gen-GKB, at the  $k$ -th step the SPR method reduces the original problem to a  $k$ -dimensional standard linear least squares problem. We further design an efficient procedure that can quickly update the iterative solution  $\mathbf{x}_k$  without solving the  $k$ -dimensional problem directly; the residual norm  $\|\mathbf{A}\mathbf{x}_k - \mathbf{b}\|_{M^{-1}}$  and solution norm  $\|\mathbf{x}_k\|_{N^{-1}}$  can also be updated quickly without computing  $M^{-1}$  and  $N^{-1}$ . Using the residual norm or solution norm, we design three early stopping rules based on DP, LC and GCV, respectively. The most computationally intensive operations in the proposed algorithm only involve matrix-vector, thereby it is very efficient for large-scale problems.
- Several theoretical results related to the regularization properties of the proposed algorithm are established. Using the generalized singular value decomposition (GSVD) of  $\{L_M \mathbf{A}, L_N\}$ , we give an explicit expression of the solution subspace and build connections between it and the best solution subspace for the SPR method (i.e. the subspace spanned by the dominant right generalized singular vectors). We prove that the iterative solution has a filtered GSVD expansion form, where some dominant GSVD components are captured and the others are filtered out. All these results reveal that the solution subspace can incorporate the information about the Gaussian noise  $\epsilon$  and desired property of  $\mathbf{x}$  encoded by the Gaussian prior, thereby the proposed algorithm can get a good regularized solution with similar accuracy as the best Tikhonov solution to (1.2).

We use both small-scale and large-scale inverse problems to test the proposed algorithm and compare it with the generalized hybrid method in Chung and Saibaba [19]. The experimental results show that our algorithm is more robust and efficient for large-scale Bayesian linear inverse problems.

This paper is organized as follows. In Section 2, we give the expression of the solution to (1.2) by using GSVD of  $\{L_M \mathbf{A}, L_N\}$ . In Section 3, we derive the gen-GKB process by treating  $\mathbf{A}$  as a bounded linear operator between the two Hilbert spaces  $(\mathbb{R}^n, \langle \cdot, \cdot \rangle_{N^{-1}})$  and  $(\mathbb{R}^m, \langle \cdot, \cdot \rangle_{M^{-1}})$ , and then propose the gen-GKB based SPR algorithm with proper early stopping rules. In Section 4, we establish several theoretical results to reveal the regularization properties of the proposed algorithm. Numerical results are presented in Section 5, and conclusions are provided in Section 6.

## 2. Solving the Bayesian linear inverse problem by GSVD

For analyzing regularized inverse problems, GSVD is a useful tool [25]. Since (1.3) and (1.2) are equivalent, we use the GSVD of the matrix pair  $\{L_M \mathbf{A}, L_N\}$  to discuss some properties of the MAP estimator that is the solution to (1.3). Since  $L_N$  is invertible, the GSVD of  $\{L_M \mathbf{A}, L_N\}$  can be written as

$$L_M \mathbf{A} = U_A \Sigma_A Z^{-1}, \quad L_N = U_L \Sigma_L Z^{-1}, \quad (2.1)$$

with

$$\Sigma_A = \begin{pmatrix} D_A & & \\ & 0 & \\ & & \end{pmatrix} \begin{matrix} r \\ m-r \\ n-r \end{matrix}, \quad \Sigma_L = \begin{pmatrix} D_L & & \\ & I & \\ & & \end{pmatrix} \begin{matrix} r \\ n-r \\ n-r \end{matrix}, \quad (2.2)$$

where  $D_A = \text{diag}(\sigma_1, \dots, \sigma_r)$ ,  $1 > \sigma_1 \geq \dots \geq \sigma_r > 0$  and  $D_L = \text{diag}(\mu_1, \dots, \mu_r)$ ,  $0 < \mu_1 \leq \dots \leq \mu_r < 1$ , and it holds that  $\sigma_i^2 + \mu_i^2 = 1$ . Throughout the paper, we denote by  $I$  and  $\mathbf{0}$  the identity matrix and zero matrix/vector, respectively, with orders clear from the context. The two matrices  $U_A \in \mathbb{R}^{m \times m}$  and  $U_L \in \mathbb{R}^{n \times n}$  are orthogonal, and  $Z$  is invertible. Let  $Z = (\mathbf{z}_1, \dots, \mathbf{z}_n)$ . Define  $\gamma_i = \sigma_i / \mu_i$  for  $1 \leq i \leq r$  and  $\gamma_i = 0$  for  $r+1 \leq i \leq n$ . We call  $\gamma_i$  the  $i$ -th largest generalized singular value and  $\mathbf{z}_i$  the  $i$ -th leading right generalized singular vector. Note that

$$Z^T (\mathbf{A}^T M^{-1} \mathbf{A}) Z = \Sigma_A^T \Sigma_A, \quad Z^T N^{-1} Z = \Sigma_L^2. \quad (2.3)$$

Therefore,  $\gamma_i^2$  are generalized eigenvalues of the matrix pair  $\{\mathbf{A}^T M^{-1} \mathbf{A}, N^{-1}\}$  with  $\mathbf{z}_i$  the corresponding generalized eigenvectors [26, §8.7]. Since  $N^{-1}$  is positive definite, we can always normalize  $\mathbf{z}_i$  such that they are mutually  $N^{-1}$ -orthonormal, i.e.  $\langle \mathbf{z}_i, \mathbf{z}_j \rangle_{N^{-1}} = \delta_{ij}$ , where  $\delta_{ij}$  is the Kronecker symbol and  $\langle \mathbf{z}, \mathbf{z}' \rangle_{N^{-1}} := \mathbf{z}^T N^{-1} \mathbf{z}'$  is the  $N^{-1}$ -inner product defined in  $\mathbb{R}^n$ . In the following part, we always assume that  $\mathbf{z}_i$  are mutually  $N^{-1}$ -orthonormal. Note that in this case,  $\Sigma_L = I$  and  $\gamma_i = \sigma_i$  for  $1 \leq i \leq r$ .

With the help of GSVD, the solution to (1.3) can be expressed as

$$\begin{aligned} \mathbf{x}_\lambda &= [(L_M \mathbf{A})^\top L_M \mathbf{A} + \lambda L_N^\top L_N]^{-1} (L_M \mathbf{A})^\top L_M \mathbf{b} \\ &= \sum_{i=1}^r \frac{\gamma_i^2}{\gamma_i^2 + \lambda} \frac{\mathbf{u}_{A,i}^\top L_M \mathbf{b}}{\sigma_i} \mathbf{z}_i \end{aligned} \quad (2.4)$$

where  $\mathbf{u}_{A,i}$  is the  $i$ -th column of  $U_A$ . This is just the filtered GSVD expansion form solution for the MAP estimator  $\mathbf{x}_\lambda$ , where the filter factors  $f_i := \gamma_i^2 / (\gamma_i^2 + \lambda)$  are applied to noisy coefficients  $\mathbf{u}_{A,i}^\top L_M \mathbf{b}$  to dampen noisy components appeared in the regularized solution [25]. The discrete Picard condition (DPC) plays a central role in the regularization of linear inverse problems, which says that the Fourier coefficients  $|\mathbf{u}_{A,i}^\top \mathbf{b}_{\text{true}}|$  on average decay to zero faster than the corresponding  $\gamma_i$ , where  $\mathbf{b}_{\text{true}} = \mathbf{A} \mathbf{x}_{\text{true}}$  is the true observation [27]. Based on DPC, the regularization parameter  $\lambda$  should be carefully chosen such that  $f_i \approx 1$  for small  $i$  and  $f_i \approx 0$  for large  $i$ .

To avoid determining  $\lambda$  in advance, the most natural alternative approach is to directly let  $f_i = 1$  for small index and  $f_i = 0$  for large index, which leads to the so called truncated GSVD (TGSVD) solution

$$\mathbf{x}_k^{\text{tgsvd}} = \sum_{i=1}^k \frac{\mathbf{u}_{A,i}^\top L_M \mathbf{b}}{\sigma_i} \mathbf{z}_i. \quad (2.5)$$

By truncating the TGSVD solution at a proper  $k$ , the obtained solution can capture main information corresponding to dominant right generalized singular vectors while suppressing noise corresponding to others [25]. These dominant  $\mathbf{z}_i$  play an important role in the regularized solution, since they contain the desirable information about  $\mathbf{x}$  encoded by the Gaussian prior.

For large-scale Bayesian inverse problems, the Cholesky factorizations of  $\mathbf{M}^{-1}$  and  $\mathbf{N}^{-1}$  and the GSVD are very hard to compute. In this case, the generalized hybrid iterative method proposed in Chung and Saibaba [19] is a good choice, which first transforms (1.2) to (1.4) and then uses gen-GKB to project (1.2) onto low dimensional subspaces to get small-scale problems. Although this method is effective for many types of inverse problems, sometimes the regularization parameter  $\lambda_k$  can not be well determined, resulting in that  $\lambda_k$  does not converge to  $\lambda_{\text{opt}}$  or converges very slowly, and the corresponding iterative solution can only have a poor accuracy.

In the following part, we approach the Bayesian linear inverse problem from the perspective of subspace projection regularization instead of Tikhonov regularization. This perspective provides a direct motivation to derive our new iterative process for generating solution subspaces. Based on this iterative process, we propose the new iterative regularization algorithm.

### 3. Subspace projection regularization based on generalized Golub-Kahan bidiagonalization

The *subspace projection regularization* (SPR) is a unified framework for developing iterative regularization methods [24, §3.3]. For the Bayesian linear inverse problem (1.2), the core idea is to compute a series of  $\mathbf{x}_k$  as the solution to

$$\min_{\mathbf{x} \in S_k} \|\mathbf{x}\|_{N^{-1}}, \quad S_k = \{\mathbf{x} : \min_{\mathbf{x} \in S_k} \|\mathbf{A}\mathbf{x} - \mathbf{b}\|_{M^{-1}}\}, \quad (3.1)$$

where  $S_k$  is the subspace of  $\mathbb{R}^n$  of dimension  $k = 1, 2, \dots$ , and the iteration should be early stopped to overcome containing too many noisy components into the solution.

**Remark 3.1.** The TGSVD solution  $\mathbf{x}_k^{\text{tgsvd}}$  is the  $k$ -th SPR solution if we choose  $S_k = \text{span}\{\mathbf{Z}_k\}$  for  $k = 1, 2, \dots, r$ , where  $\mathbf{Z}_k = (\mathbf{z}_1, \dots, \mathbf{z}_k)$ . Throughout the paper, we denote by  $\text{span}\{\cdot\}$  the subspace spanned by columns of a matrix or a group of vectors. To see it, let  $\mathbf{x} = \mathbf{Z}_k \mathbf{y}$  to denote any vector in  $\text{span}\{\mathbf{Z}_k\}$ , where  $\mathbf{y} \in \mathbb{R}^k$ . Then we have

$$\begin{aligned} \min_{\mathbf{x} \in S_k} \|\mathbf{A}\mathbf{x} - \mathbf{b}\|_{M^{-1}} &= \min_{\mathbf{x} = \mathbf{Z}_k \mathbf{y}} \|\mathbf{L}_M (\mathbf{A}\mathbf{x} - \mathbf{b})\|_2 \\ &= \min_{\mathbf{y} \in \mathbb{R}^k} \|\mathbf{L}_M \mathbf{A} \mathbf{Z}_k \mathbf{y} - \mathbf{L}_M \mathbf{b}\|_2 \\ &= \min_{\mathbf{y} \in \mathbb{R}^k} \|\mathbf{U}_{A,k} \boldsymbol{\Sigma}_{A,k} \mathbf{y} - \mathbf{L}_M \mathbf{b}\|_2, \end{aligned}$$

where  $\mathbf{U}_{A,k}$  is the first  $k$  part of  $\mathbf{U}_A$  and  $\boldsymbol{\Sigma}_{A,k}$  is the first  $k$ -by- $k$  part of  $\boldsymbol{\Sigma}$ , and the last equality uses the GSVD relation (2.1). The above least squares problem has a unique solution  $\mathbf{y}_k = \boldsymbol{\Sigma}_{A,k}^{-1} \mathbf{U}_{A,k}^\top \mathbf{L}_M \mathbf{b}$ . Thus, the solution to (3.1) is  $\mathbf{x}_k = \mathbf{Z}_k \mathbf{y}_k = \sum_{i=1}^k \frac{\mathbf{u}_{A,i}^\top L_M \mathbf{b}}{\sigma_i} \mathbf{z}_i$ , which is just  $\mathbf{x}_k^{\text{tgsvd}}$ .

In fact, for any choice of  $S_k$ , there exists a unique solution to (3.1). Using the same method as the above, let  $\mathbf{x} = \mathbf{W}_k \mathbf{y}$  be any vector in  $S_k$ , where  $\mathbf{W}_k \in \mathbb{R}^{n \times k}$  whose columns are  $N^{-1}$ -orthonormal and  $\text{span } S_k$ . Then the solution to (3.1) satisfies  $\mathbf{x}_k = \mathbf{W}_k \mathbf{y}_k$ , where  $\mathbf{y}_k$  is the solution to

$$\min_{\mathbf{y} \in \mathbb{R}^k} \|\mathbf{y}\|_2, \quad \mathcal{Y}_k = \{\mathbf{x} : \min_{\mathbf{y} \in \mathbb{R}^k} \|\mathbf{L}_M \mathbf{A} \mathbf{W}_k \mathbf{y} - \mathbf{L}_M \mathbf{b}\|_2\}.$$

This problem has a unique solution  $\mathbf{y}_k = (\mathbf{L}_M \mathbf{A} \mathbf{W}_k)^\dagger \mathbf{L}_M \mathbf{b}$ , where  $C^\dagger$  is the pseudoinverse of a matrix  $C$ . Therefore, there exists a unique  $\mathbf{x}_k$ .

Comparing to computing the MAP estimator  $\mathbf{x}_\lambda$  in the Tikhonov regularization framework (1.2), the SPR framework utilizes those elaborately constructed subspaces  $S_k$  to enforce additional structure on  $\mathbf{x}$ , which plays a similar role as the Tikhonov regularization term  $\lambda \|\mathbf{x}\|_{N^{-1}}$  arising from the Gaussian prior of  $\mathbf{x}$ . We call  $S_k$  the *solution subspace*, which should be constructed such that the prior information about the desired solution is incorporated into it. Based on the expressions of  $\mathbf{x}_\lambda$  and  $\mathbf{x}_k^{\text{tgsvd}}$ , the most ideal choice is  $S_k = \text{span}\{\mathbf{Z}_k\}$ , which just leads to the TGSVD solution.

### 3.1. Generalized Golub-Kahan bidiagonalization

For large-scale problems, computing GSVD is impractical. A reasonable SPR method should construct a series of solution subspaces that capture main information corresponding to some leading right generalized singular vectors  $\{z_i\}_{i=1}^k$  as the iteration proceeds. Note that  $z_i$  are mutually  $N^{-1}$ -orthonormal, and note that (3.1) has the form of weighted least squares. This motivates us to design an iterative process that generates a series of  $N^{-1}$ -orthonormal vectors and projects (3.1) onto subspaces spanned by these vectors.

**Definition 3.1.** Define two finite dimensional Hilbert spaces as  $(\mathbb{R}^n, \langle \cdot, \cdot \rangle_{N^{-1}})$  and  $(\mathbb{R}^m, \langle \cdot, \cdot \rangle_{M^{-1}})$  that are equipped with the  $N^{-1}$ -inner product  $\langle x, x' \rangle_{N^{-1}} := x^\top N^{-1} x'$  and  $M^{-1}$ -inner product  $\langle y, y' \rangle_{M^{-1}} := y^\top M^{-1} y'$ , respectively.

Although the two topologies of  $N^{-1}$ - and standard inner product spaces are equivalent, the geometry structures of these two spaces are different. For dealing with (3.1), using  $N^{-1}$ - and  $M^{-1}$ -inner product is more natural. In order to develop iterative process for  $A$  between the two spaces  $(\mathbb{R}^n, \langle \cdot, \cdot \rangle_{N^{-1}})$  and  $(\mathbb{R}^m, \langle \cdot, \cdot \rangle_{M^{-1}})$ , we first need the adjoint of the linear operator  $A$ . For the linear operator  $A : (\mathbb{R}^n, \langle \cdot, \cdot \rangle_{N^{-1}}) \rightarrow (\mathbb{R}^m, \langle \cdot, \cdot \rangle_{M^{-1}})$  defined as  $A : x \mapsto Ax$  for  $x \in \mathbb{R}^n$  under the canonical bases of  $\mathbb{R}^n$  and  $\mathbb{R}^m$ , define its adjoint  $A^* : (\mathbb{R}^m, \langle \cdot, \cdot \rangle_{M^{-1}}) \rightarrow (\mathbb{R}^n, \langle \cdot, \cdot \rangle_{N^{-1}})$  as  $A^* : y \mapsto A^*y$  for  $y \in \mathbb{R}^m$ , by the relation

$$\langle Ax, y \rangle_{M^{-1}} = \langle x, A^*y \rangle_{N^{-1}}.$$

Since the two Hilbert spaces are finite dimensional, it follows that  $A$  is bounded, and the adjoint operator  $A^*$  is a well-defined linear operator. Under the above two definitions, we can develop an iterative process that reduces  $A$  to a bidiagonal form and generates two subspaces of  $(\mathbb{R}^n, \langle \cdot, \cdot \rangle_{N^{-1}})$  and  $(\mathbb{R}^m, \langle \cdot, \cdot \rangle_{M^{-1}})$ , respectively. This can be achieved by applying the GKB process to  $A$  with starting vector  $b$  between the two Hilbert spaces  $(\mathbb{R}^n, \langle \cdot, \cdot \rangle_{N^{-1}})$  and  $(\mathbb{R}^m, \langle \cdot, \cdot \rangle_{M^{-1}})$ ; for the GKB process for compact linear operator, see Caruso and Novati [28]. The basic recursive relations are as follows:

$$\beta_1 u_1 = b, \quad (3.2)$$

$$\alpha_i v_i = A^* u_i - \beta_i v_{i-1}, \quad (3.3)$$

$$\beta_{i+1} u_{i+1} = A v_i - \alpha_i u_i, \quad (3.4)$$

where  $\alpha_i$  and  $\beta_i$  are computed such that  $\|u_i\|_{M^{-1}} = \|v_i\|_{N^{-1}} = 1$ , and we let  $v_0 = 0$ . In order to compute  $A^* u_i$ , we need the matrix-form expression of  $A^*$ .

**Lemma 3.1.** Under the canonical bases of  $\mathbb{R}^n$  and  $\mathbb{R}^m$ , we have

$$A^* = N A^\top M^{-1}. \quad (3.5)$$

**Proof.** Under the canonical bases, the linear mapping  $Ax$  and  $A^*y$  are the standard matrix-vector products. By the definition of  $A^*$ , we have

$$(Ax)^\top M^{-1} y = x^\top N^{-1} A^* y$$

for any  $x \in \mathbb{R}^n$  and  $y \in \mathbb{R}^m$ . Therefore we have  $A^\top M^{-1} = N^{-1} A^*$ , and the desired result is immediately obtained.  $\square$

In order to avoid explicitly computing  $N^{-1}$ -norm to obtain  $\alpha_i$ , let  $\bar{u}_i = M^{-1} u_i$  and  $\bar{v}_i = N^{-1} v_i$ . By (3.3) and (3.5), we have

$$\alpha_i \bar{v}_i = A^\top \bar{u}_i - \beta_i \bar{v}_{i-1}. \quad (3.6)$$

Let  $\bar{r}_i = A^\top \bar{u}_i - \beta_i \bar{v}_{i-1}$  and  $s_i = A v_i - \alpha_i u_i$ . Then  $\alpha_i = \|N \bar{r}_i\|_{N^{-1}} = (\bar{r}_i^\top N \bar{r}_i)^{1/2}$  and  $\beta_{i+1} = \|s_i\|_{M^{-1}} = (s_i^\top M^{-1} s_i)^{1/2}$ . We remark that computing with  $M^{-1}$  can not be avoided to get  $u_i$ . For the most commonly encountered cases that  $\epsilon$  is a Gaussian white noise or all of its components are uncorrelated,  $M$  is diagonal and  $M^{-1}$  can be directly obtained. The whole iterative process is summarized in Algorithm 1.

**Algorithm 1** Generalized Golub-Kahan bidiagonalization (gen-GKB).

**Require:**  $A \in \mathbb{R}^{m \times n}$ ,  $b \in \mathbb{R}^m$ ,  $M \in \mathbb{R}^{m \times m}$ ,  $N \in \mathbb{R}^{n \times n}$

```

1:  $\bar{s} = M^{-1} b$ 
2:  $\beta_1 = \bar{s}^\top \bar{s}$ ,  $u_1 = \bar{s} / \beta_1$ ,  $\bar{u}_1 = \bar{s} / \beta_1$ 
3:  $\bar{r} = A^\top \bar{u}_1$ ,  $r = N \bar{r}$ 
4:  $\alpha_1 = (r^\top r)^{1/2}$ ,  $\bar{v}_1 = \bar{r} / \alpha_1$ ,  $v_1 = r / \alpha_1$ 
5: for  $i = 1, 2, \dots, k$  do
6:    $s = A v_i - \alpha_i u_i$ 
7:    $\bar{s} = M^{-1} s$ 
8:    $\beta_{i+1} = (\bar{s}^\top \bar{s})^{1/2}$ ,  $u_{i+1} = \bar{s} / \beta_{i+1}$ ,  $\bar{u}_{i+1} = \bar{s} / \beta_{i+1}$   $\triangleright \bar{u}_{i+1} = M^{-1} u_{i+1}$ 
9:    $\bar{r} = A^\top \bar{u}_{i+1} - \beta_{i+1} \bar{v}_i$ 
10:   $r = N \bar{r}$ 
11:   $\alpha_{i+1} = (r^\top r)^{1/2}$ ,  $\bar{v}_{i+1} = \bar{r} / \alpha_{i+1}$ ,  $v_{i+1} = r / \alpha_{i+1}$   $\triangleright \bar{v}_{i+1} = N^{-1} v_{i+1}$ 
12: end for
Ensure:  $\{\alpha_i, \beta_i\}_{i=1}^{k+1}$ ,  $\{u_i, v_i\}_{i=1}^{k+1}$ ,  $\{\bar{u}_i, \bar{v}_i\}_{i=1}^{k+1}$ 
```

One can verify that the gen-GKB derived here is equivalent to the iterative process used in Chung and Saibaba [19] with the same name. However, here we adopt a new point of view, i.e. treating  $A$  as a linear operator between two Hilbert spaces  $(\mathbb{R}^n, \langle \cdot, \cdot \rangle_{N^{-1}})$  and  $(\mathbb{R}^m, \langle \cdot, \cdot \rangle_{M^{-1}})$  and applying GKB to  $A$  between these two spaces to get gen-GKB. This view is particularly intuitive for constructing valid solution subspaces in SPR methods for the Bayesian inverse problems. Based on this new view, we gain insight into the structures of the two subspaces  $\text{span}\{u_i\}_{i=1}^k$  and  $\text{span}\{v_i\}_{i=1}^k$ .

**Proposition 3.1.** *The group of vectors  $\{u_i\}_{i=1}^k$  is an  $M^{-1}$ -orthonormal basis of the Krylov subspace*

$$\mathcal{K}_k(ANA^T M^{-1}, b) = \text{span}\{(ANA^T M^{-1})^i b\}_{i=0}^{k-1},$$

and  $\{v_i\}_{i=1}^k$  is an  $N^{-1}$ -orthonormal basis of the Krylov subspace

$$\mathcal{K}_k(NA^T M^{-1}A, NA^T M^{-1}b) = \text{span}\{(NA^T M^{-1}A)^i NA^T M^{-1}b\}_{i=0}^{k-1}.$$

**Proof.** We exploit the property of GKB for the linear operator  $A$  with starting vector  $b$  between the two Hilbert spaces  $(\mathbb{R}^n, \langle \cdot, \cdot \rangle_{N^{-1}})$  and  $(\mathbb{R}^m, \langle \cdot, \cdot \rangle_{M^{-1}})$ , which states that  $\{u_i\}_{i=1}^k$  and  $\{v_i\}_{i=1}^k$  are the  $M^{-1}$ -orthonormal basis and  $N^{-1}$ -orthonormal basis of the Krylov subspaces  $\mathcal{K}_k(AA^*, b)$  and  $\mathcal{K}_k(A^*A, A^*b)$ , respectively. Using Lemma 3.1, we have  $AA^* = ANA^T M^{-1}$ ,  $A^*A = NA^T M^{-1}A$  and  $A^*b = NA^T M^{-1}b$ . The proof is completed.  $\square$

Define two matrices  $U_k = (u_1, \dots, u_k)$  and  $V_k = (v_1, \dots, v_k)$ . Then  $U_k$  and  $V_k$  are two  $M^{-1}$  and  $N^{-1}$  orthonormal matrices, respectively. By (3.2)–(3.4), we can rewrite gen-GKB with the matrix-form recursive relations:

$$\beta_1 U_{k+1} e_1 = b, \quad (3.7)$$

$$AV_k = U_{k+1} B_k, \quad (3.8)$$

$$NA^T M^{-1} U_{k+1} = V_k B_k^T + \alpha_{k+1} v_{k+1} e_{k+1}^T, \quad (3.9)$$

where  $e_1$  and  $e_{k+1}$  are the first and  $(k+1)$ -th columns of the identity matrix of order  $k+1$ , respectively, and

$$B_k = \begin{pmatrix} \alpha_1 & & & & \\ \beta_2 & \alpha_2 & & & \\ & \beta_3 & \ddots & & \\ & & \ddots & \ddots & \\ & & & \ddots & \alpha_k \\ & & & & \beta_{k+1} \end{pmatrix} \in \mathbb{R}^{(k+1) \times k} \quad (3.10)$$

is the projection of  $A$  onto subspaces  $\text{span}\{U_{k+1}\}$  and  $\text{span}\{V_k\}$ . If gen-GKB does not terminate in  $k$  steps, i.e.  $\alpha_i, \beta_i \neq 0$  for  $1 \leq i \leq k+1$ , then  $B_k$  has full column rank. Based on the above matrix-form relations, we can design the iterative SPR algorithm based on gen-GKB.

### 3.2. Iterative regularization by subspace projection

At the  $k$ -th iteration of gen-GKB, we seek solution to (3.1) by setting  $S_k = \text{span}\{V_k\}$ . Denote any vector  $x \in S_k$  as  $x = V_k y$  with  $y \in \mathbb{R}^k$ . By (3.8), we have

$$\begin{aligned} \min_{x \in S_k} \|Ax - b\|_{M^{-1}} &= \min_{y \in \mathbb{R}^k} \|AV_k y - U_{k+1} \beta_1 e_1\|_{M^{-1}} \\ &= \min_{y \in \mathbb{R}^k} \|U_{k+1} (B_k y - U_{k+1} \beta_1 e_1)\|_{M^{-1}} \\ &= \min_{y \in \mathbb{R}^k} \|B_k y - \beta_1 e_1\|_2, \end{aligned}$$

where we use the property that  $U_{k+1}$  is  $M^{-1}$ -orthonormal. Under the assumption that gen-GKB does not terminate, the above  $k$ -dimensional linear least squares problem has a unique solution. Therefore, the solution to (3.1) is

$$x_k = V_k y_k, \quad y_k = \underset{y \in \mathbb{R}^k}{\text{argmin}} \|B_k y - \beta_1 e_1\|_2 \quad (3.11)$$

for  $k = 1, 2, \dots$ . Furthermore, we do not need to explicitly compute  $y_k$  by solving (3.11) for each  $k$ . Instead, there is an efficient procedure to update  $x_k$  from  $x_0 = 0$  step by step. The procedure is similar to the LSQR algorithm for standard linear least squares problems [29].

First, perform the following QR factorization:

$$\tilde{Q}_k (B_k \quad \beta_1 e_1) = \begin{pmatrix} R_k & g_k \\ & \tilde{\phi}_{k+1} \end{pmatrix} = \begin{pmatrix} \rho_1 & \theta_2 & & & & \phi_1 \\ & \rho_2 & \theta_3 & & & \phi_2 \\ & & \ddots & & & \vdots \\ & & & \rho_{k-1} & \theta_k & \phi_{k-1} \\ - & - & - & - & \rho_k & \phi_k \\ & & & & & \tilde{\phi}_{k+1} \end{pmatrix},$$

where the orthogonal matrix  $\tilde{Q}_k$  is the product of a series of Givens rotation matrices that is used to zero out  $\beta_{i+1}$ :

$$\begin{pmatrix} c_i & s_i \\ s_i & -c_i \end{pmatrix} \begin{pmatrix} \bar{\rho}_i & 0 \\ \beta_{i+1} & \alpha_{i+1} \end{pmatrix} \begin{pmatrix} \bar{\phi}_i \\ 0 \end{pmatrix} = \begin{pmatrix} \rho_i & \theta_{i+1} & \phi_i \\ 0 & \bar{\rho}_{i+1} & \bar{\phi}_{i+1} \end{pmatrix},$$

where  $i = 1, \dots, k$ . There exist efficient recursive formulae to compute these matrix elements, and here we adopt these formulae proposed in Paige and Saunders [29] (line 3–6 in Algorithm 2). Using the orthogonality of  $\tilde{\mathbf{Q}}_k$  leads to

$$\|\mathbf{B}_k \mathbf{y} - \beta_1 \mathbf{e}_1\|_2^2 = \left\| \tilde{\mathbf{Q}}_k (\mathbf{B}_k \quad \beta_1 \mathbf{e}_1) \begin{pmatrix} \mathbf{y} \\ -1 \end{pmatrix} \right\|_2^2 = \|\mathbf{R}_k \mathbf{y} - \mathbf{g}_k\|_2^2 + |\bar{\phi}_{k+1}|^2. \quad (3.12)$$

Thereby the solution to  $\operatorname{argmin}_{\mathbf{y} \in \mathbb{R}^k} \|\mathbf{B}_k \mathbf{y} - \beta_1 \mathbf{e}_1\|_2$  is  $\mathbf{y}_k = \mathbf{R}_k^{-1} \mathbf{g}_k$ . Factorizing  $\mathbf{R}_k$  as

$$\mathbf{R}_k = \mathbf{D}_k \bar{\mathbf{R}}_k, \quad \mathbf{D}_k = \begin{pmatrix} \rho_1 & & & \\ & \rho_2 & & \\ & & \ddots & \\ & & & \rho_k \end{pmatrix}, \quad \bar{\mathbf{R}}_k = \begin{pmatrix} 1 & \theta_2/\rho_1 & & \\ & 1 & \theta_3/\rho_2 & \\ & & \ddots & \theta_k/\rho_{k-1} \\ & & & 1 \end{pmatrix},$$

we obtain

$$\mathbf{x}_k = \mathbf{V}_k \mathbf{y}_k = \mathbf{V}_k \mathbf{R}_k^{-1} \mathbf{g}_k = (\mathbf{V}_k \bar{\mathbf{R}}_k^{-1})(\mathbf{D}_k^{-1} \mathbf{g}_k). \quad (3.13)$$

Let  $\mathbf{W}_k = \mathbf{V}_k \bar{\mathbf{R}}_k^{-1} = (\mathbf{w}_1, \dots, \mathbf{w}_k)$ . Note that  $\mathbf{D}_k^{-1} \mathbf{g}_k = (\phi_1/\rho_1, \dots, \phi_k/\rho_k)^\top$ . Solving  $\mathbf{W}_k \bar{\mathbf{R}}_k = \mathbf{V}_k$  by back substitution and combining (3.13) leads to the recursive formula for  $\mathbf{x}_i$  and  $\mathbf{w}_i$ :

$$\mathbf{x}_i = \mathbf{x}_{i-1} + (\phi_i/\rho_i) \mathbf{w}_i, \quad \mathbf{w}_{i+1} = \mathbf{v}_{i+1} - (\theta_{i+1}/\rho_i) \mathbf{w}_i. \quad (3.14)$$

From the above derivation, we have

$$\bar{\phi}_{k+1} = \|\mathbf{B}_k \mathbf{y}_k - \beta_1 \mathbf{e}_1\|_2 = \|\mathbf{A} \mathbf{x}_k - \mathbf{b}\|_{\mathbf{M}^{-1}}. \quad (3.15)$$

Thereby the residual norm can be updated very quickly without explicitly computing  $\|\mathbf{A} \mathbf{x}_k - \mathbf{b}\|_{\mathbf{M}^{-1}}$ . In order to compute  $\|\mathbf{x}_k\|_{\mathbf{N}^{-1}}$  without explicitly using  $\mathbf{N}^{-1}$ , let  $\bar{\mathbf{x}}_k = \mathbf{N}^{-1} \mathbf{x}_k$  and  $\bar{\mathbf{w}}_k = \mathbf{N}^{-1} \mathbf{w}_k$ . By (3.14) we have

$$\bar{\mathbf{x}}_i = \bar{\mathbf{x}}_{i-1} + (\phi_i/\rho_i) \bar{\mathbf{w}}_i, \quad \bar{\mathbf{w}}_{i+1} = \bar{\mathbf{v}}_{i+1} - (\theta_{i+1}/\rho_i) \bar{\mathbf{w}}_i, \quad (3.16)$$

and  $\|\mathbf{x}_k\|_{\mathbf{N}^{-1}} = (\mathbf{x}_k^\top \bar{\mathbf{x}}_k)^{1/2}$ . The above updating procedure is summarized in Algorithm 2.

---

**Algorithm 2** Procedure for updating iterative solutions and residual/solution norms.

---

```

1: Let  $\mathbf{x}_0 = \mathbf{0}$ ,  $\bar{\mathbf{x}}_0 = \mathbf{0}$ ,  $\mathbf{w}_1 = \mathbf{v}_1$ ,  $\bar{\mathbf{w}}_1 = \bar{\mathbf{v}}_1$ ,  $\bar{\phi}_1 = \beta_1$ ,  $\bar{\rho}_1 = \alpha_1$ 
2: for  $i = 1, 2, \dots, k$  do
3:    $\rho_i = (\bar{\rho}_i^2 + \beta_{i+1}^2)^{1/2}$ 
4:    $c_i = \bar{\rho}_i/\rho_i$ ,  $s_i = \beta_{i+1}/\rho_i$ 
5:    $\theta_{i+1} = s_i \alpha_{i+1}$ ,  $\bar{\rho}_{i+1} = -c_i \alpha_{i+1}$ 
6:    $\phi_i = c_i \bar{\phi}_i$ ,  $\bar{\phi}_{i+1} = s_i \bar{\phi}_i$ 
7:    $\mathbf{x}_i = \mathbf{x}_{i-1} + (\phi_i/\rho_i) \mathbf{w}_i$ 
8:    $\mathbf{w}_{i+1} = \mathbf{v}_{i+1} - (\theta_{i+1}/\rho_i) \mathbf{w}_i$ 
9:    $\bar{\mathbf{x}}_i = \bar{\mathbf{x}}_{i-1} + (\phi_i/\rho_i) \bar{\mathbf{w}}_i$ 
10:   $\bar{\mathbf{w}}_{i+1} = \bar{\mathbf{v}}_{i+1} - (\theta_{i+1}/\rho_i) \bar{\mathbf{w}}_i$ 
11:   $\|\mathbf{x}_k\|_{\mathbf{N}^{-1}} = (\mathbf{x}_k^\top \bar{\mathbf{x}}_k)^{1/2}$ 
12: end for
```

---

### 3.3. Early stopping rules

The SPR method usually exhibits semi-convergence behavior: as the iteration proceeds, the iterative solution first gradually approximate to  $\mathbf{x}_{\text{true}}$ , then the solution will deviate far from  $\mathbf{x}_{\text{true}}$  and eventually converges to  $\mathbf{x}_{\text{naive}} = \mathbf{A}^\dagger \mathbf{b}$  [24, §3.3]. This is because the solution subspace will contain more and more noisy components as it gradually expands. The iteration at which the corresponding solution has the smallest error is called the semi-convergence point. In order to obtain a good regularized solution, we should stop the iteration early. Note that the iteration number  $k$  in SPR plays a similar role as the regularization parameter in Tikhonov regularization. Here we adapt several criteria for choosing regularization parameters to estimate the semi-convergence point.

**Discrepancy principle.** For the case that  $\epsilon$  is a Gaussian white noise, if  $\|\epsilon\|_2$  is known, one criterion for determining the early stopping iteration is the discrepancy principle (DP), which states that the discrepancy between the data and predicted output  $\|\mathbf{A} \mathbf{x}_k - \mathbf{b}\|_2$ , should be of the order of  $\|\epsilon\|_2$  [30, §7.2]. If  $\epsilon$  is a non-white Gaussian noise, noticing that (1.1) leads to

$$\mathbf{L}_M \mathbf{b} = \mathbf{L}_M \mathbf{A} \mathbf{x} + \mathbf{L}_M \epsilon, \quad (3.17)$$

and  $\mathbf{L}_M \epsilon \sim \mathcal{N}(\mathbf{0}, \mathbf{I})$ , thereby this transformation whitens the noise, and the discrepancy is  $\|\mathbf{L}_M \mathbf{A} \mathbf{x}_k - \mathbf{L}_M \mathbf{b}\|_2 = \|\mathbf{A} \mathbf{x}_k - \mathbf{b}\|_{\mathbf{M}^{-1}}$ . Since  $\bar{\epsilon} = \mathbf{L}_M \epsilon$  is a Gaussian white noise with zero mean, it follows that

$$\mathbb{E}[\|\bar{\epsilon}\|_2^2] = \mathbb{E}[\operatorname{trace}(\bar{\epsilon}^\top \bar{\epsilon})] = \mathbb{E}[\operatorname{trace}(\bar{\epsilon} \bar{\epsilon}^\top)] = \operatorname{trace}(\mathbb{E}[\bar{\epsilon} \bar{\epsilon}^\top]) = \operatorname{trace}(\mathbf{I}) = m.$$

Note from (3.15) that  $\|\mathbf{A}\mathbf{x}_k - \mathbf{b}\|_{\mathbf{M}^{-1}} = \bar{\phi}_{k+1}$  decreases monotonically since  $\mathbf{x}_k$  minimizes  $\|\mathbf{A}\mathbf{x} - \mathbf{b}\|_{\mathbf{M}^{-1}}$  in the gradually expanding subspace  $\text{span}\{\mathbf{V}_k\}$ . Following DP, we should stop iteration at the first  $k$  satisfying

$$\bar{\phi}_{k+1} \leq \tau \sqrt{m} \quad (3.18)$$

with  $\tau > 1$  slightly, such as  $\tau = 1.01$ . Typically, the early stopping iteration determined by DP is slightly smaller than the semi-convergence point, thereby the corresponding solution is over-regularized.

*L-curve.* Another early stopping rule is the L-curve criterion, which is a heuristic approach that does not need the noise norm [9]. The idea is to plot in log-log scale the following curve

$$(\log \|\mathbf{A}\mathbf{x}_k - \mathbf{b}\|_{\mathbf{M}^{-1}}, \log \|\mathbf{x}_k\|_{\mathbf{N}^{-1}}) = (\log \bar{\phi}_{k+1}, \log(\mathbf{x}_k^\top \bar{\mathbf{x}}_k)^{1/2}), \quad (3.19)$$

which frequently has a characteristic 'L' shape, and the corner corresponds to the point where further reduction in the residual comes only at the expense of a drastic increase in the regularization term. Thus, we should choose the early stopping iteration that corresponds to this corner, which is usually defined as the point of maximum curvature of the L-curve in a log-log plot.

*Generalized cross-validation.* The generalized cross-validation (GCV) is a statistical approach for estimating the optimal regularization parameter, especially for the Gaussian white noise case [10]. For the SPR method where the iteration number  $k$  plays the role of the regularization parameter, this approach can be adapted to estimate the optimal iteration number  $k_0$ . Since the transformed system (3.17) equivalent to (1.1) has a white Gaussian noise  $\mathbf{L}_M \epsilon$ , the GCV function with respect to  $k$  should be

$$\text{GCV}(k) = \frac{\|\mathbf{L}_M \mathbf{A}\mathbf{x}_k - \mathbf{L}_M \mathbf{b}\|_2^2}{(\text{trace}(\mathbf{I} - \mathbf{L}_M \mathbf{A} \mathbf{A}_k^\dagger))^2} = \frac{\|\mathbf{A}\mathbf{x}_k - \mathbf{b}\|_{\mathbf{M}^{-1}}^2}{(\text{trace}(\mathbf{I} - \mathbf{L}_M \mathbf{A} \mathbf{A}_k^\dagger))^2},$$

where  $\mathbf{A}_k^\dagger$  denotes the matrix that maps the right-hand side  $\mathbf{L}_M \mathbf{b}$  to the  $k$ -th regularized solution  $\mathbf{x}_k$ . By (3.11) and (3.7) we have

$$\mathbf{x}_k = \mathbf{V}_k \mathbf{B}_k^\dagger \beta_1 \mathbf{e}_1 = \mathbf{V}_k \mathbf{B}_k^\dagger \mathbf{U}_{k+1}^\top \mathbf{M}^{-1} \mathbf{b} = (\mathbf{V}_k \mathbf{B}_k^\dagger \mathbf{U}_{k+1}^\top \mathbf{M}^{-1} \mathbf{L}_M^{-1}) \mathbf{L}_M \mathbf{b},$$

which implies that  $\mathbf{A}_k^\dagger = \mathbf{V}_k \mathbf{B}_k^\dagger \mathbf{U}_{k+1}^\top \mathbf{M}^{-1} \mathbf{L}_M^{-1}$ . Using (3.8) again we get

$$\begin{aligned} \text{trace}(\mathbf{L}_M \mathbf{A} \mathbf{A}_k^\dagger) &= \text{trace}(\mathbf{L}_M \mathbf{A} \mathbf{V}_k \mathbf{B}_k^\dagger \mathbf{U}_{k+1}^\top \mathbf{M}^{-1} \mathbf{L}_M^{-1}) = \text{trace}(\mathbf{U}_{k+1} \mathbf{B}_k \mathbf{B}_k^\dagger \mathbf{U}_{k+1}^\top \mathbf{M}^{-1}) \\ &= \text{trace}(\mathbf{B}_k \mathbf{B}_k^\dagger \mathbf{U}_{k+1}^\top \mathbf{M}^{-1} \mathbf{U}_{k+1}) \\ &= \text{trace}(\mathbf{B}_k \mathbf{B}_k^\dagger) = k. \end{aligned}$$

Therefore, the final expression of the GCV function is

$$\text{GCV}(k) = \frac{\bar{\phi}_{k+1}^2}{(m-k)^2}, \quad (3.20)$$

and the minimizer of this function is used as a good estimate of  $k_0$ .

We remark that in using the GCV or L-curve method, one must go a few iterations beyond the optimal  $k$  in order to verify the optimum. The gen-GKB based SPR iterative algorithm for (1.1) is summarized in Algorithm 3.

---

**Algorithm 3** Subspace projection regularization by gen-GKB (genGKB\_SPR).

---

**Require:**  $\mathbf{A} \in \mathbb{R}^{m \times n}$ ,  $\mathbf{b} \in \mathbb{R}^m$ ,  $\mathbf{M} \in \mathbb{R}^{m \times m}$ ,  $\mathbf{N} \in \mathbb{R}^{n \times n}$

- 1: Initialize  $\mathbf{x}_0 = \mathbf{0}$ ,  $\bar{\mathbf{x}}_0 = \mathbf{0}$ ,
- 2: **for**  $k = 1, 2, \dots$ , **do**
- 3:   Compute  $\mathbf{v}_k$ ,  $\bar{\mathbf{v}}_k$ ,  $\alpha_k$ ,  $\beta_k$  by gen-GKB
- 4:   Compute  $\mathbf{x}_k$ ,  $\bar{\mathbf{x}}_k$ ,  $\bar{\phi}_{k+1}$ ,  $\|\mathbf{x}_k\|_{\mathbf{N}^{-1}}$  by Algorithm 1
- 5:   **if** Early stopping criterion is satisfied **then**
- 6:     Denote the estimated optimal iteration by  $k_1$
- 7:     Terminate the iteration at  $k_1$
- 8:   **end if**
- 9: **end for**

**Ensure:** Final regularized solution  $\mathbf{x}_{k_1}$

---

#### 4. Regularization properties of genGKB\_SPR

We establish several theoretical results about the genGKB\_SPR algorithm to reveal its regularization properties for solving the Bayesian linear inverse problem. Based on the GSVD of  $\{\mathbf{L}_M \mathbf{A}, \mathbf{L}_N\}$  appeared from the transformed standard 2-norm Tikhonov regularization (1.3), we first give an explicit expression of the solution subspaces  $\text{span}\{\mathbf{V}_k\}$  and build connections with  $\text{span}\{\mathbf{Z}_k\}$ , which are the most ideal solution subspaces for the SPR method.

**Lemma 4.1.** Following the notations in (2.1) and (2.2), the  $k$ -th solution subspace of  $\text{genGKB\_SPR}$  is

$$\text{span}\{\mathbf{V}_k\} = \text{span}\{\mathbf{Z}_r \mathbf{D}_A^{2i+1} \mathbf{U}_{A,r}^\top \mathbf{L}_M \mathbf{b}\}_{i=0}^{k-1}, \quad (4.1)$$

where  $\mathbf{Z}_r$  and  $\mathbf{U}_{A,r}$  are the first columns of  $\mathbf{Z}$  and  $\mathbf{U}_A$ , respectively.

**Proof.** By Proposition 3.1 we have  $\text{span}\{\mathbf{V}_k\} = \mathcal{K}_k(\mathbf{N} \mathbf{A}^\top \mathbf{M}^{-1} \mathbf{A}, \mathbf{N} \mathbf{A}^\top \mathbf{M}^{-1} \mathbf{b})$ . Using the GSVD of  $\{\mathbf{L}_M \mathbf{A}, \mathbf{L}_N\}$ , we have

$$\mathbf{N} \mathbf{A}^\top \mathbf{M}^{-1} \mathbf{A} = \mathbf{N} \mathbf{Z}^{-\top} \boldsymbol{\Sigma}_A^\top \mathbf{U}_A^\top \mathbf{L}_M^{-\top} \mathbf{M}^{-1} \mathbf{L}_M^{-1} \mathbf{U}_A \boldsymbol{\Sigma}_A \mathbf{Z}^{-1} = \mathbf{Z} \boldsymbol{\Sigma}_A^\top \boldsymbol{\Sigma}_A \mathbf{Z}^{-1},$$

where we use  $\mathbf{N} \mathbf{Z}^{-\top} = \mathbf{Z}$  since  $\mathbf{Z}^\top \mathbf{N}^{-1} \mathbf{Z} = \mathbf{I}$ . Similarly,

$$\mathbf{N} \mathbf{A}^\top \mathbf{M}^{-1} \mathbf{b} = \mathbf{N} \mathbf{Z}^{-\top} \boldsymbol{\Sigma}_A^\top \mathbf{U}_A^\top \mathbf{L}_M^{-\top} \mathbf{M}^{-1} \mathbf{b} = \mathbf{Z} \boldsymbol{\Sigma}_A^\top \mathbf{U}_A^\top \mathbf{L}_M \mathbf{b}.$$

Therefore, for  $i = 0, 1, \dots, k-1$ , it holds

$$\begin{aligned} (\mathbf{N} \mathbf{A}^\top \mathbf{M}^{-1} \mathbf{A})^i \mathbf{N} \mathbf{A}^\top \mathbf{M}^{-1} \mathbf{b} &= (\mathbf{Z} \boldsymbol{\Sigma}_A^\top \boldsymbol{\Sigma}_A \mathbf{Z}^{-1})^i \mathbf{Z} \boldsymbol{\Sigma}_A^\top \mathbf{U}_A^\top \mathbf{L}_M \mathbf{b} \\ &= \mathbf{Z} (\boldsymbol{\Sigma}_A^\top \boldsymbol{\Sigma}_A)^i \boldsymbol{\Sigma}_A^\top \mathbf{U}_A^\top \mathbf{L}_M \mathbf{b} \\ &= \mathbf{Z}_r \mathbf{D}_A^{2i+1} \mathbf{U}_{A,r}^\top \mathbf{L}_M \mathbf{b}. \end{aligned}$$

The desired result immediately follows.  $\square$

To investigate how far the  $k$ -th solution subspace is from the ideal one  $\text{span}\{\mathbf{Z}_k\}$ , we use the following definition for measuring the distance between two subspaces under a non-standard inner product; see Stewart [31, Chapter 4, §2.1] and Björck [32, §2.2.5] for the corresponding definition under the standard inner product.

**Definition 4.1.** Let  $\mathcal{F}$  and  $\mathcal{G}$  be two  $k$ -dimensional subspaces of  $(\mathbb{R}^n, \langle \cdot, \cdot \rangle_{N^{-1}})$ . Define the maximum canonical angle between  $\mathcal{F}$  and  $\mathcal{G}$  as

$$\Theta(\mathcal{F}, \mathcal{G}) = \max_{\substack{\mathbf{u} \in \mathcal{F} \\ \mathbf{u} \neq 0}} \min_{\substack{\mathbf{v} \in \mathcal{G} \\ \mathbf{v} \neq 0}} \arccos \frac{|\langle \mathbf{u}, \mathbf{v} \rangle_{N^{-1}}|}{\|\mathbf{u}\|_{N^{-1}} \|\mathbf{v}\|_{N^{-1}}}.$$

The distance between  $\mathcal{F}$  and  $\mathcal{G}$  is defined as

$$\text{dist}(\mathcal{F}, \mathcal{G}) = \sin \Theta(\mathcal{F}, \mathcal{G}).$$

**Lemma 4.2.** Let  $(\mathbf{F}_1, \mathbf{F}_2)$  and  $(\mathbf{G}_1, \mathbf{G}_2)$  be two  $N^{-1}$ -orthogonal matrices, both of whose columns span  $(\mathbb{R}^n, \langle \cdot, \cdot \rangle_{N^{-1}})$ . If  $\text{span}\{\mathbf{F}_1\} = \mathcal{F}$  and  $\text{span}\{\mathbf{G}_1\} = \mathcal{G}$ , then

$$\text{dist}(\mathcal{F}, \mathcal{G}) = \|\mathbf{F}_2^\top \mathbf{N}^{-1} \mathbf{G}_1\|_2. \quad (4.2)$$

**Proof.** Several matrix-form expressions of  $\Theta(\mathcal{F}, \mathcal{G})$  have been studied in Knyazev and Argentati [33, Section 4]. Here we exploit Knyazev and Argentati [33, Theorem 4.2], which states that the canonical angles between  $\mathcal{F}$  and  $\mathcal{G}$  under  $N^{-1}$ -inner product coincide with the canonical angles between  $\mathbf{L}_N \mathcal{F}$  and  $\mathbf{L}_N \mathcal{G}$  under 2-inner product. Let  $\bar{\Theta}(\mathcal{F}, \mathcal{G})$  be the maximum canonical angle between  $\mathbf{L}_N \mathcal{F}$  and  $\mathbf{L}_N \mathcal{G}$  under 2-inner product. By Stewart [31, Theorem 2.2] we have

$$\text{dist}(\mathcal{F}, \mathcal{G}) = \sin \bar{\Theta}(\mathcal{F}, \mathcal{G}) = \|(\mathbf{L}_N \mathbf{F}_2)^\top (\mathbf{L}_N \mathbf{G}_1)\|_2 = \|\mathbf{F}_2^\top \mathbf{N}^{-1} \mathbf{G}_1\|_2.$$

The first “=” uses the fact that  $(\mathbf{L}_N \mathbf{F}_1, \mathbf{L}_N \mathbf{F}_2)$  and  $(\mathbf{L}_N \mathbf{G}_1, \mathbf{L}_N \mathbf{G}_2)$  are two orthogonal matrices, and  $\text{span}\{\mathbf{L}_N \mathbf{F}_1\} = \mathbf{L}_N \mathcal{F}$  and  $\text{span}\{\mathbf{L}_N \mathbf{G}_1\} = \mathbf{L}_N \mathcal{G}$ .  $\square$

Now we establish an explicit expression of the distance between  $\text{span}\{\mathbf{V}_k\}$  and  $\text{span}\{\mathbf{Z}_k\}$ . To this end, we exploit the following commonly used DPC model for (1.3):

$$|\mathbf{u}_{A,i}^\top \mathbf{L}_M \mathbf{b}_{\text{true}}| = \gamma_i^{1+\beta}, \quad i = 1, \dots, r, \quad (4.3)$$

where  $\beta > 0$  controls the decay rate of the Fourier coefficients  $\mathbf{u}_{A,i}^\top \mathbf{L}_M \mathbf{b}_{\text{true}}$  with respect to  $\gamma_i$  [34, §4.6]. Note that  $\mathbf{L}_M \mathbf{b} = \mathbf{L}_M \mathbf{b}_{\text{true}} + \mathbf{L}_M \epsilon$  and  $\mathbf{L}_M \epsilon$  is a Gaussian white noise. Therefore, it follows that  $\mathbf{u}_{A,i}^\top \mathbf{L}_M \mathbf{b} = \mathbf{u}_{A,i}^\top \mathbf{L}_M \mathbf{b}_{\text{true}} + \mathbf{u}_{A,i}^\top \mathbf{L}_M \epsilon$ , where  $\tilde{\epsilon} = \mathbf{U}_{A,r}^\top \mathbf{L}_M \epsilon$  is a Gaussian white noise. By DPC,  $\mathbf{u}_{A,i}^\top \mathbf{L}_M \mathbf{b}_{\text{true}}$  dominates in  $\mathbf{u}_{A,i}^\top \mathbf{L}_M \mathbf{b}$  for small  $i$  while  $\mathbf{u}_{A,i}^\top \mathbf{L}_M \epsilon$  dominates for big  $i$ , thereby it is reasonable to assume that  $\mathbf{u}_{A,i}^\top \mathbf{L}_M \mathbf{b}$  is nonzero for all  $1 \leq i \leq r$ . Based on the above model, we have the following result.

**Theorem 4.1.** Suppose  $\{\gamma_i\}_{i=1}^r$  have distinct values. For any  $1 \leq k \leq r$ , let

$$\begin{aligned} \mathbf{D} = \text{diag}(\mathbf{D}_A \mathbf{U}_{A,r}^\top \mathbf{L}_M \mathbf{b}) &= \begin{pmatrix} \mathbf{D}_1 & & \\ & \mathbf{D}_2 & \\ & & \ddots \end{pmatrix} \begin{matrix} k \\ r-k \\ r-k \end{matrix}, \\ \mathbf{H}_k &= \begin{pmatrix} 1 & \sigma_1^2 & \dots & \sigma_{1k-2}^{2k-2} \\ 1 & \sigma_2^2 & \dots & \sigma_{2k-2}^{2k-2} \\ \vdots & \vdots & \ddots & \vdots \\ 1 & \sigma_r^2 & \dots & \sigma_r^{2k-2} \end{pmatrix} = \begin{pmatrix} \mathbf{H}_{k1} \\ \mathbf{H}_{k2} \\ \vdots \end{pmatrix} \begin{matrix} k \\ r-k \\ r-k \end{matrix}, \end{aligned}$$

and

$$\Delta_k = D_2 H_{k2} H_{k1}^{-1} D_1^{-1} \in \mathbb{R}^{(r-k) \times k}. \quad (4.4)$$

Then the distance between the  $k$ -th solution subspace  $S_k = \text{span}\{\mathbf{V}_k\}$  and  $\mathcal{Z}_k = \text{span}\{\mathbf{Z}_k\}$  is

$$\text{dist}(S_k, \mathcal{Z}_k) = \frac{\|\Delta_k\|_2}{(1 + \|\Delta_k\|_2^2)^{1/2}}. \quad (4.5)$$

**Proof.** First notice that the diagonals of  $D_1$  are nonzero and the Vandermonde matrix  $H_{k1}$  is invertible since  $\{\sigma_i\}_{i=1}^r = \{\gamma_i\}_{i=1}^r$  have distinct values. Lemma 4.1 implies  $S_k = \text{span}\{\mathbf{Z}_r D H_k\}$ . Since

$$\mathbf{Z}_r D H_k = \mathbf{Z}_r \begin{pmatrix} D_1 H_{k1} \\ D_2 H_{k2} \end{pmatrix} = \mathbf{Z}_r \begin{pmatrix} I \\ \Delta_k \end{pmatrix} D_1 H_{k1},$$

it follows that  $S_k = \text{span}\left\{\mathbf{Z}_r \begin{pmatrix} I \\ \Delta_k \end{pmatrix}\right\}$ . Using the relation

$$\left(\mathbf{Z}_r \begin{pmatrix} I \\ \Delta_k \end{pmatrix}\right)^\top N^{-1} \mathbf{Z}_r \begin{pmatrix} I \\ \Delta_k \end{pmatrix} = I + \Delta_k^\top \Delta_k,$$

one can check that the columns of  $\mathbf{Z}_r \begin{pmatrix} I \\ \Delta_k \end{pmatrix} (I + \Delta_k^\top \Delta_k)^{-1/2}$  constitute an  $N^{-1}$ -orthonormal basis of  $S_k$ . Partition  $\mathbf{Z}$  as  $\mathbf{Z} =$

$$\begin{pmatrix} \mathbf{Z}_k & \mathbf{Z}_{k1} & \mathbf{Z}_{k2} \\ k & r-k & n-r \end{pmatrix}. \text{ By Lemma 4.2, we get}$$

$$\begin{aligned} \text{dist}(S_k, \mathcal{Z}_k) &= \left\| \begin{pmatrix} \mathbf{Z}_{k1} & \mathbf{Z}_{k2} \end{pmatrix}^\top N^{-1} \begin{pmatrix} \mathbf{Z}_k & \mathbf{Z}_{k1} \end{pmatrix} \begin{pmatrix} I \\ \Delta_k \end{pmatrix} (I + \Delta_k^\top \Delta_k)^{-1/2} \right\|_2 \\ &= \left\| \begin{pmatrix} \Delta_k \\ \mathbf{0} \end{pmatrix} (I + \Delta_k^\top \Delta_k)^{-1/2} \right\|_2 = \frac{\|\Delta_k\|_2}{(1 + \|\Delta_k\|_2^2)^{1/2}}. \end{aligned}$$

The proof is completed.  $\square$

Theorem 4.1 generalizes the results in Hansen [30, §6.4] under a similar model and assumption. The value of  $\|\Delta_k\|_2$  is essential for measuring the distance between  $S_k$  and  $\mathcal{Z}_k$ , where a qualitative investigation is presented in Hansen [30, §6.4]. Here we gain more insight into the variation of  $\|\Delta_k\|_2$  as  $k$  increases. First, we have

$$\begin{aligned} \|\Delta_k\|_2 &\leq \|D_2\|_2 \left\| H_{k2} H_{k1}^{-1} \right\|_2 \left\| D_1^{-1} \right\|_2 \\ &= \frac{\sigma_{k+1}}{\sigma_k} \frac{\max_{k+1 \leq i \leq r} |\mathbf{u}_{A,i}^\top \mathbf{L}_M \mathbf{b}|}{\min_{1 \leq i \leq k} |\mathbf{u}_{A,i}^\top \mathbf{L}_M \mathbf{b}|} \left\| H_{k2} H_{k1}^{-1} \right\|_2. \end{aligned} \quad (4.6)$$

Following the proof of Hansen [30, Theorem 6.4.1], the  $i$ -th column of  $H_{k2} H_{k1}^{-1}$  is  $\left( L_i^{(k)}(\sigma_{k+1}^2), \dots, L_i^{(k)}(\sigma_r^2) \right)^\top$ , where  $L_i^{(k)}(\lambda) = \prod_{j=1, j \neq i}^k (\lambda - \sigma_j^2) / (\sigma_i^2 - \sigma_j^2)$  for  $1 \leq i \leq k$ ,  $k > 1$ , and  $L_1^{(1)}(\lambda) \equiv 1$ . Notice that  $|L_i^{(k)}(\lambda)|$  is monotonically decreasing for  $\lambda \in [0, \sigma_k^2]$ . The upper bound on it is  $|L_i^{(k)}(0)|$ . Let  $|L_{i_0}^{(k)}(0)| = \max_{1 \leq i \leq k} |L_i^{(k)}(0)|$ . Then we obtain

$$\begin{aligned} \left\| H_{k2} H_{k1}^{-1} \right\|_2 &\leq \sqrt{k} \max_{1 \leq i \leq k} \left\| [H_{k2} H_{k1}^{-1}]_i \right\|_2 \\ &\leq \sqrt{k} \max_{1 \leq i \leq k} \sqrt{r-k} |L_i^{(k)}(0)| = \sqrt{k(r-k)} |L_{i_0}^{(k)}(0)|, \end{aligned} \quad (4.7)$$

where  $[\cdot]_i$  denotes the  $i$ -th column of a matrix.

To further investigate the upper bound on  $\|\Delta_k\|_2$  in (4.6), we use the simplified model  $\sigma_i = \gamma_i = \gamma_0 e^{-\alpha i}$  for  $i = 1, \dots, r$  to describe the decaying rate of  $\gamma_i$ , i.e. they decay exponentially<sup>1</sup> For small  $k$  that the noise components in  $\mathbf{u}_{A,i}^\top \mathbf{L}_M \mathbf{b}$  do not dominate, using the model (4.3) we have  $\frac{\sigma_{k+1}}{\sigma_k} \frac{\max_{k+1 \leq i \leq r} |\mathbf{u}_{A,i}^\top \mathbf{L}_M \mathbf{b}|}{\min_{1 \leq i \leq k} |\mathbf{u}_{A,i}^\top \mathbf{L}_M \mathbf{b}|} \approx (\gamma_{k+1}/\gamma_k)^{2+\beta} = e^{-\alpha(2+\beta)}$ . Then we estimate the bound on  $\left\| H_{k2} H_{k1}^{-1} \right\|_2$  in (4.7). Obviously  $|L_{i_0}^{(1)}(0)| = 1$ . For  $k \geq 2$ , we have

$$|L_{i_0}^{(k)}(0)| = \prod_{j=1, j \neq i_0}^k \left| \frac{\sigma_j^2}{\sigma_j^2 - \sigma_{i_0}^2} \right| = \prod_{j=1}^{i_0-1} \frac{\sigma_j^2}{\sigma_j^2 - \sigma_{i_0}^2} \prod_{j=i_0+1}^k \frac{\sigma_j^2}{\sigma_{i_0}^2 - \sigma_j^2}$$

<sup>1</sup> This model describes the so called “severely ill-posed” forward operator  $\mathbf{A}$ . Another model  $\gamma_i = \gamma_1 i^{-\alpha}$  for  $i = 1, \dots, r$  describes the “modestly/mildly ill-posed”  $\mathbf{A}$ ; see Engl et al. [24, §2.2] for more discussions and examples.

$$\begin{aligned}
&= \prod_{j=1}^{i_0-1} \frac{1}{1 - e^{-2(i_0-j)\alpha}} \prod_{j=i_0+1}^k \frac{1}{e^{2(j-i_0)\alpha} - 1} \\
&= \prod_{j=1}^{i_0-1} \frac{1}{1 - e^{-2j\alpha}} \prod_{j=1}^{k-i_0} \frac{1}{1 - e^{-2j\alpha}} \prod_{j=1}^{k-i_0} \frac{1}{e^{2j\alpha}} \\
&\leq \prod_{j=1}^{i_0-1} \left(1 + \frac{1}{2j\alpha}\right) \prod_{j=1}^{k-i_0} \left(1 + \frac{1}{2j\alpha}\right) e^{-\alpha(k-i_0)(k-i_0+1)},
\end{aligned}$$

where we use  $1/(1 - e^{-x}) \leq 1 + 1/x$  for  $x > -1$ . Notice that for  $i_0 = 1$  the first term vanishes, and for  $i_0 = k$  the second and third terms vanish. Define  $S_1 = \prod_{j=1}^{i_0-1} \left(1 + \frac{1}{2j\alpha}\right)$ . Then

$$\ln S_1 = \sum_{j=1}^{i_0-1} \ln \left(1 + \frac{1}{2j\alpha}\right) \leq \frac{1}{2\alpha} \sum_{j=1}^{i_0-1} \frac{1}{j} \leq \frac{1}{2\alpha} \left(1 + \int_1^{i_0-1} \frac{1}{j} dx\right) = \frac{1 + \ln(i_0 - 1)}{2\alpha}.$$

Similarly, we have  $\ln \prod_{j=1}^{k-i_0} \left(1 + \frac{1}{2j\alpha}\right) \leq \frac{1 + \ln(k-i_0)}{2\alpha}$ . Therefore, we obtain

$$\begin{aligned}
\ln |L_{i_0}^{(k)}(0)| &\leq \frac{1}{2\alpha} (2 + \ln(i_0 - 1)(k - i_0)) - \alpha(k - i_0)(k - i_0 + 1) \\
&\leq \frac{1}{2\alpha} \left(2 + \ln \frac{(k-1)^2}{4}\right) - \alpha(k - i_0)(k - i_0 + 1),
\end{aligned}$$

which leads to

$$|L_{i_0}^{(k)}(0)| \leq \left(\frac{e(k-1)}{2}\right)^{\frac{1}{\alpha}} e^{-\alpha(k-i_0)(k-i_0+1)}.$$

If  $i_0$  keeps around a constant value when  $k$  increases, the above bound is a gradually decreasing quantity. If  $k - i_0$  always remains a constant or decreases when  $k$  increases, the above bound is pessimistic since it increases at the speed of  $k^{1/\alpha}$ . In this case, we derive an asymptotic bound:

$$\begin{aligned}
|L_{i_0}^{(k)}(0)| &\leq \prod_{j=1}^{i_0-1} \frac{1}{1 - e^{-2j\alpha}} = \prod_{j=1}^{i_0-1} (1 + \mathcal{O}(e^{-2j\alpha})) \\
&= 1 + \mathcal{O}\left(\sum_{j=1}^{i_0-1} e^{-2j\alpha}\right) = 1 + \mathcal{O}\left(\frac{e^{-2\alpha}}{1 - e^{-2\alpha}}\right).
\end{aligned}$$

A sharp upper bound on  $|L_{i_0}^{(k)}(0)|$  is extremely difficult since  $i_0$  may vary for different  $k$ , but in the worst case it grows from 1 very slightly controlled by  $\mathcal{O}(e^{-2\alpha})$ .

From the above derivation, we have

$$\|\Delta_k\|_2 \lesssim \sqrt{k(r-k)} e^{-\alpha(2+\beta)} |L_{i_0}^{(k)}(0)| \quad (4.8)$$

for small  $k$  that the noise in  $|\mathbf{u}_{A,k}^\top \mathbf{L}_M \mathbf{b}|$  is negligible. The bounded factor  $\sqrt{k(r-k)}$  is an overestimate that can be reduced more or less, and  $|L_{i_0}^{(k)}(0)|$  is slightly larger than 1 in the worst case. Therefore, we can loosely speak that  $\|\Delta_k\|_2$  remains small before the noise dominates, which implies that  $S_k$  captures main information about  $\mathbf{x}$  contained in  $\mathcal{Z}_k$ . Going further, we have the following result.

**Theorem 4.2.** Let the compact singular value decomposition (SVD) of  $\mathbf{B}_k$  be  $\mathbf{B}_k = \mathbf{J}_k \mathbf{\Gamma}_k \mathbf{P}_k^\top$ , where  $\mathbf{\Gamma}_k = \text{diag}(\theta_1^{(k)}, \dots, \theta_k^{(k)})$  with  $\theta_1^{(k)} > \theta_2^{(k)} > \dots > \theta_k^{(k)} > 0$  and  $\mathbf{P}_k = (\mathbf{p}_1^{(k)}, \dots, \mathbf{p}_k^{(k)})$ . Then each pair  $\left((\theta_i^{(k)})^2, \mathbf{V}_k \mathbf{p}_i^{(k)}\right)$  converges to one of the generalized eigenvalue-vector pairs of  $\mathbf{A}^\top \mathbf{M}^{-1} \mathbf{A} \mathbf{z} = \xi \mathbf{N}^{-1} \mathbf{z}$  as  $k$  increases.

**Proof.** By (3.8) and (3.9), we have

$$\mathbf{A}^\top \mathbf{M}^{-1} \mathbf{A} \mathbf{V}_k = (\mathbf{A}^\top \mathbf{M}^{-1} \mathbf{U}_{k+1}) \mathbf{B}_k = \mathbf{N}^{-1} \mathbf{V}_k \mathbf{B}_k^\top \mathbf{B}_k + \alpha_{k+1} \beta_{k+1} \mathbf{N}^{-1} \mathbf{v}_{k+1} \mathbf{e}_k^\top,$$

where  $\mathbf{e}_k$  is the  $k$ -th column of the identity matrix of order  $k$ . By (3.3) and (3.7), we have

$$\alpha_1 \beta_1 \mathbf{v}_1 = \mathbf{N} \mathbf{A}^\top \mathbf{M}^{-1} \beta_1 \mathbf{u}_1 = \mathbf{N} \mathbf{A}^\top \mathbf{M}^{-1} \mathbf{b}.$$

Notice that  $\mathbf{B}_k^\top \mathbf{B}_k$  is a symmetric tridiagonal matrix with diagonals  $\alpha_i^2 + \beta_{i+1}^2$  and subdiagonals  $\alpha_i \beta_i$  for  $i = 1, \dots, k$ . The above two relations indicate that  $\mathbf{T}_k$  is the Ritz-Galerkin projection of  $\mathbf{A}^\top \mathbf{M}^{-1} \mathbf{A}$  onto subspace  $\text{span}\{\mathbf{V}_k\}$  generated by the Lanczos tridiagonalization of  $\mathbf{A}^\top \mathbf{M}^{-1} \mathbf{A}$  with starting vector  $\mathbf{N} \mathbf{A}^\top \mathbf{M}^{-1} \mathbf{b}$  under  $\mathbf{N}^{-1}$ -inner product [35, §5.5]. Since the eigenvalue decomposition of  $\mathbf{B}_k^\top \mathbf{B}_k$  is  $\mathbf{B}_k^\top \mathbf{B}_k = \mathbf{P}_k \mathbf{\Gamma}_k^2 \mathbf{P}_k^\top$ , by the convergence theory of Lanczos tridiagonalization for generalized eigenvalue problems, we obtain the desired result.  $\square$

By (2.3), the generalized eigenvalue-vector pairs for  $A^T M^{-1} A z = \xi N^{-1} z$  are  $\{(\gamma_i^2, z_i)\}_{i=1}^n$ . The convergence behavior of  $((\theta_i^{(k)})^2, V_k P_i^{(k)})$  as  $k$  increases is governed by the Kaniel-Paige-Saad theory that is similar to the eigenvalue problem of a single matrix [26, §10.1.5], which states that it has a faster convergence to those generalized eigenvalues (and corresponding eigenvectors) lying in the two ends of the spectrum and well separated. Since  $\gamma_i^2$  decreases and clusters at zero, we can hope for a faster convergence of  $((\theta_i^{(k)})^2, V_k P_i^{(k)})$  to some largest generalized eigenvalue and corresponding vector. Therefore, the information in  $S_k$  can be extracted to approximate dominant generalized eigenvectors  $z_i$ . These vectors constitute basic features about  $x$  encoded by the Gaussian prior. In fact, the regularized solution computed by genGKB\_SPR contains those dominant  $z_i$  while filtering out others. To reveal it, we need the following lemma, which generalizes [36, Property 2.8] for the conjugate gradient (CG) method to a rank-deficient matrix.

**Lemma 4.3.** For a symmetric positive semi-definite linear system  $Cf = d$  with  $d \in \mathcal{R}(C)$ , where  $\mathcal{R}(\cdot)$  is the range space of a matrix, the  $k$ -th CG solution of it with  $d_0 = 0$  can be written as  $f_k = (I - \mathcal{R}_k(C))C^\dagger d$ , where  $\mathcal{R}_k$  is the normalized characteristic polynomial of  $T_k$  as (4.11) such that  $\mathcal{R}_k(0) = 1$ .

**Proof.** Suppose the  $k$ -step standard Lanczos tridiagonalization of  $C$  with starting vector  $d$  is written as the following matrix-form:

$$CQ_k = Q_k T_k + \delta_{k+1} q_{k+1} e_k^T, \quad (4.9)$$

$$\delta_1 q_1 = d, \quad (4.10)$$

where  $\{q_i\}_{i=1}^{k+1}$  are orthonormal and

$$T_k = \begin{pmatrix} \eta_1 & \delta_2 & & \\ \delta_2 & \eta_2 & \ddots & \\ & \ddots & \ddots & \delta_k \\ & & \delta_k & \eta_k \end{pmatrix}. \quad (4.11)$$

We note that this process must terminate, i.e.  $\delta_{k+1} = 0$  for at most  $k = l$  steps, where  $l = \text{rank}(C)$ , and the corresponding solution is just  $C^\dagger d$ ; if the process does not terminate at the  $k$ -th step, then  $T_k$  is invertible [32, §4.2.3–§4.2.5]. To prove this lemma, we use the fact that the  $k$ -th CG solution of  $Cf = d$  with  $d_0 = 0$  is  $f_k = Q_k T_k^{-1} \delta_1 e_1$ .

Let the compact SVD of  $C$  be  $C = V_C \Sigma_C V_C^T$ , where  $\Sigma_C \in \mathbb{R}^{l \times l}$  has positive diagonals and  $V_C$  has  $l$  orthonormal columns. Since  $d \in \mathcal{R}(C) = \mathcal{R}(V_C)$ , it follows that  $q_i \in \mathcal{R}(V_C)$  for  $i = 1, \dots, k+1$ . Let  $q_i = V_C \bar{q}_i$  with  $\bar{q}_i \in \mathbb{R}^l$  and  $\bar{Q}_k = (\bar{q}_1, \dots, \bar{q}_k)$ . It follows that  $\{\bar{q}_i\}_{i=1}^{k+1}$  are orthonormal. From (4.9) and (4.10) we get

$$\begin{aligned} \Sigma_C \bar{Q}_k &= \bar{Q}_k T_k + \delta_{k+1} \bar{q}_{k+1} e_k^T, \\ \delta_1 \bar{q}_1 &= V_C^T d, \end{aligned}$$

which is just the matrix-form of the Lanczos tridiagonalization of  $\Sigma_C$  with starting vector  $V_C^T d$ . Let  $\bar{f}_k = \bar{Q}_k T_k^{-1} \delta_1 e_1$ . Since  $\Sigma_C$  is positive definite, by Van der Sluis and van der Vorst [36, Property 2.8] we have  $\bar{f}_k = (I - \mathcal{R}_k(\Sigma_C)) \Sigma_C^{-1} V_C^T d$ . Using  $Q_k = V_C \bar{Q}_k$ , we obtain

$$\begin{aligned} f_k &= V_C \bar{f}_k = V_C (I - \mathcal{R}_k(\Sigma_C)) \Sigma_C^{-1} V_C^T d \\ &= (I - \mathcal{R}_k(V_C \Sigma_C V_C^T)) V_C \Sigma_C^{-1} V_C^T d \\ &= (I - \mathcal{R}_k(C)) C^\dagger d. \end{aligned}$$

The proof is completed.  $\square$

**Theorem 4.3.** The  $k$ -th regularized solution computed by genGKB\_SPR has the expression

$$x_k = \sum_{i=1}^r f_i^{(k)} \frac{u_{A,i}^T L_M b}{\sigma_i} z_i \quad (4.12)$$

with filter factors

$$f_i^{(k)} = 1 - \prod_{j=1}^k \frac{(\theta_j^{(k)})^2 - \gamma_i^2}{(\theta_j^{(k)})^2}, \quad i = 1, \dots, r. \quad (4.13)$$

**Proof.** Following the proof of Theorem 4.2, we have

$$\begin{aligned} (L_N^{-T} A^T M^{-1} A L_N^{-1}) L_N V_k &= L_N V_k (B_k^T B_k) + \alpha_k \beta_k L_N v_{k+1} e_k, \\ \alpha_1 \beta_1 L_N v_1 &= L_N^{-T} A^T M b, \end{aligned}$$

and  $\{L_N v_i\}_{i=1}^{k+1}$  are 2-orthonormal. This implies that  $L_N v_i$  are the Lanczos vectors generated by the standard Lanczos tridiagonalization of  $L_N^{-T} A^T M^{-1} A L_N^{-1}$  with starting vector  $L_N^{-T} A^T M b$ . Let  $\tilde{x}_k = L_N x_k$ . Then by (3.11) we have  $\tilde{x}_k = L_N V_k (B_k^T B_k)^{-1} B_k^T \beta_1 e_1 = (L_N V_k) T_k^{-1} \alpha_1 \beta_1 e_1$ . Therefore,  $\tilde{x}_k$  is the  $k$ -th CG solution of the symmetric positive semi-definite linear system

$$L_N^{-T} A^T M^{-1} A L_N^{-1} x = L_N^{-T} A^T M b. \quad (4.14)$$

Note that

$$\mathbf{L}_N^{-\top} \mathbf{A}^\top \mathbf{M} \mathbf{b} = (\mathbf{L}_M \mathbf{A} \mathbf{L}_N^{-1})^\top \mathbf{L}_M \mathbf{b}, \quad \mathbf{L}_N^{-\top} \mathbf{A}^\top \mathbf{M}^{-1} \mathbf{A} \mathbf{L}_N^{-1} = (\mathbf{L}_M \mathbf{A} \mathbf{L}_N^{-1})^\top (\mathbf{L}_M \mathbf{A} \mathbf{L}_N^{-1}).$$

It follows that  $\mathbf{L}_N^{-\top} \mathbf{A}^\top \mathbf{M} \mathbf{b} \in \mathcal{R}(\mathbf{L}_N^{-\top} \mathbf{A}^\top \mathbf{M}^{-1} \mathbf{A} \mathbf{L}_N^{-1})$  and

$$\begin{aligned} & (\mathbf{L}_N^{-\top} \mathbf{A}^\top \mathbf{M}^{-1} \mathbf{A} \mathbf{L}_N^{-1})^\dagger \mathbf{L}_N^{-\top} \mathbf{A}^\top \mathbf{M} \mathbf{b} \\ &= (\mathbf{L}_M \mathbf{A} \mathbf{L}_N^{-1})^\dagger \mathbf{L}_M \mathbf{b} = (\mathbf{U}_A \boldsymbol{\Sigma}_A (\mathbf{L}_N \mathbf{Z})^{-1})^\dagger \mathbf{L}_M \mathbf{b} \\ &= \mathbf{L}_N \mathbf{Z} \boldsymbol{\Sigma}_A^\dagger \mathbf{U}_A^\top \mathbf{L}_M \mathbf{b}, \end{aligned}$$

where we use (2.1) and the property that  $\mathbf{L}_N \mathbf{Z}$  has orthonormal columns. Therefore, by Lemma 4.3 we have

$$\tilde{\mathbf{x}}_k = (\mathbf{I} - \mathcal{R}_k(\mathbf{L}_N^{-\top} \mathbf{A}^\top \mathbf{M}^{-1} \mathbf{A} \mathbf{L}_N^{-1})) \mathbf{L}_N \mathbf{Z} \boldsymbol{\Sigma}_A^\dagger \mathbf{U}_A^\top \mathbf{L}_M \mathbf{b}, \quad (4.15)$$

where  $\mathcal{R}_k$  is the normalized characteristic polynomial of  $\mathbf{B}_k^\top \mathbf{B}_k$  such that  $\mathcal{R}_k(0) = 1$ .

By (2.3), we have  $\mathbf{L}_N^{-\top} \mathbf{A}^\top \mathbf{M}^{-1} \mathbf{A} \mathbf{L}_N^{-1} = \mathbf{L}_N^{-\top} \mathbf{Z}^{-\top} \boldsymbol{\Sigma}_A^\top \boldsymbol{\Sigma}_A \mathbf{Z}^{-1} \mathbf{L}_N^{-1}$ . Since  $\mathbf{Z}^{-1} \mathbf{L}_N^{-1} \mathbf{L}_N^{-\top} \mathbf{Z}^{-\top} = (\mathbf{Z}^\top \mathbf{N}^{-1} \mathbf{Z})^{-1} = \mathbf{I}$ , we get  $\mathbf{I} - \mathcal{R}_k(\mathbf{L}_N^{-\top} \mathbf{A}^\top \mathbf{M}^{-1} \mathbf{A} \mathbf{L}_N^{-1}) = \mathbf{L}_N^{-\top} \mathbf{Z}^{-\top} \boldsymbol{\Sigma}_A^\top \boldsymbol{\Sigma}_A \mathbf{Z}^{-1} \mathbf{L}_N^{-1}$ , where  $\boldsymbol{\Lambda} = \text{diag}(\{1 - \mathcal{R}_k(\sigma_i^2)\}_{i=1}^n)$ , where  $\sigma_i := 0$  for  $r+1 \leq i \leq n$ . Therefore, (4.15) leads to  $\tilde{\mathbf{x}}_k = \mathbf{L}_N^{-\top} \mathbf{Z}^{-\top} \boldsymbol{\Sigma}_A^\top \mathbf{U}_A^\top \mathbf{L}_M \mathbf{b}$ , and

$$\begin{aligned} \mathbf{x}_k &= \mathbf{L}_N^{-1} \mathbf{L}_N^{-\top} \mathbf{Z}^{-\top} \boldsymbol{\Sigma}_A^\dagger \mathbf{U}_A^\top \mathbf{L}_M \mathbf{b} = \mathbf{Z} \boldsymbol{\Lambda} \boldsymbol{\Sigma}_A^\dagger \mathbf{U}_A^\top \mathbf{L}_M \mathbf{b} \\ &= (\mathbf{Z}_r \quad \mathbf{Z}_{r1}) \begin{pmatrix} \boldsymbol{\Lambda}_r & \\ & \boldsymbol{\Lambda}_{r1} \end{pmatrix} \begin{pmatrix} \mathbf{D}_A^{-1} & \\ & \mathbf{0} \end{pmatrix} \begin{pmatrix} \mathbf{U}_{A,r}^\top \\ \mathbf{U}_{A,r1}^\top \end{pmatrix} \mathbf{L}_M \mathbf{b} \\ &= \mathbf{Z}_r \boldsymbol{\Lambda}_r \mathbf{D}_A^{-1} \mathbf{U}_{A,r}^\top \mathbf{L}_M \mathbf{b}, \end{aligned}$$

where  $\mathbf{Z}$ ,  $\boldsymbol{\Lambda}$  and  $\mathbf{U}_A$  are partitioned such that the above matrix multiplication is reasonable. By Theorem 4.2, the eigenvalues of  $\mathbf{B}_k^\top \mathbf{B}_k$  are  $\{(\theta_j^{(k)})^2\}_{j=1}^k$ . It follows that  $\mathcal{R}_k((\theta_j^{(k)})^2) = 0$  for  $1 \leq j \leq k$ . Since  $\mathcal{R}_k(0) = 1$  and the degree of  $\mathcal{R}_k$  is  $k$ , we have

$$\mathcal{R}_k(\theta) = \prod_{j=1}^k \frac{(\theta_j^{(k)})^2 - \theta}{(\theta_j^{(k)})^2}.$$

Therefore, we finally obtain  $\mathbf{x}_k = \sum_{i=1}^r f_i^{(k)} \frac{\mathbf{u}_{A,i}^\top \mathbf{L}_M \mathbf{b}}{\sigma_i} \mathbf{z}_i$  with  $f_i^{(k)}$  defined as (4.13).  $\square$

Theorem 4.3 shows that  $\mathbf{x}_k$  has a filtered GSVD expansion form. If the  $k$  Ritz values  $\{(\theta_j^{(k)})^2\}_{j=1}^k$  approximate the first  $k$  generalized singular values  $\{\gamma_i^2\}_{i=1}^k$  in natural order, i.e.  $(\theta_j^{(k)})^2 \approx \gamma_i^2$  for  $i = 1, \dots, k$ , from (4.13) we can justify that  $f_i^{(k)} \approx 1$  for  $i = 1, \dots, k$  and  $f_i^{(k)}$  decreases monotonically to zero for  $i = k+1, \dots, r$ . Comparing (4.12) with (2.4) and (2.5), we find that  $\mathbf{x}_k$  contains the first  $k$  dominant GSVD components and filters the others. Therefore, as  $k$  increases, the iterative solution first approximates  $\mathbf{x}_{\text{true}}$ , but as  $k$  becomes too big,  $\mathbf{x}_k$  will diverge from  $\mathbf{x}_{\text{true}}$  since too many noisy components are contained. The transition point is the semi-convergence point, precisely what the early stopping rules aim to estimate.

## 5. Experimental results

We conduct several numerical experiments to demonstrate the performance of our proposed algorithm. The generalized hybrid iterative method proposed in Chung and Saibaba [19] is used as a comparison, which is also based on gen-GKB and equivalent to solving (1.2) by projecting it onto the solution space  $\text{span}\{\mathbf{V}_k\}$  to get

$$\min_{\mathbf{x} \in \mathbf{V}_k} \{\|\mathbf{A}\mathbf{x} - \mathbf{b}\|_{\mathbf{M}^{-1}}^2 + \lambda \|\mathbf{x}\|_{\mathbf{N}^{-1}}^2\} = \min_{\mathbf{y} \in \mathbb{R}^k} \|\mathbf{B}_k \mathbf{y} - \beta_1 \mathbf{e}_1\|_2^2 + \lambda \|\mathbf{y}\|_2^2. \quad (5.1)$$

In order to solve the above  $k$  dimensional projected problem, at each iteration, the weighted GCV (WGCV) method [22] is adopted to determine a regularization parameter  $\lambda_k$  to form  $\lambda_k \|\mathbf{y}\|_2^2$ . In the experiments, we name the generalized hybrid iterative method with WGCV as genHyb\_WGCV. The convergence behaviors of the two methods are shown by plotting the variation of relative reconstruction error  $\|\mathbf{x}_k - \mathbf{x}_{\text{true}}\|_2 / \|\mathbf{x}_{\text{true}}\|_2$  with respect to  $k$ . The reconstructed solutions are also drawn to further compare the two methods. All the experiments are performed in MATLAB R2019b, where some codes in Hansen [37], Gazzola et al. [38] are exploited.

### 5.1. Small-scale problems

We first choose two small-scale linear inverse problems from Hansen [37] for the test. In this case, we can factorize  $\mathbf{M}^{-1}$  and  $\mathbf{N}^{-1}$  to form (1.3), and solve it directly to find the optimal Tikhonov regularization parameter  $\lambda_{\text{opt}}$  and corresponding solution  $\mathbf{x}_{\text{opt}}$ , that is  $\lambda_{\text{opt}} = \min_{\lambda > 0} \|\mathbf{x}_\lambda - \mathbf{x}_{\text{true}}\|_2$ . We use this optimal solution as a baseline for comparing the two methods.

**Experiment 1.** The first small-scale test problem is gravity, which is a one-dimensional model for gravity surveying. It aims to reconstruct a mass distribution  $f(t)$  located at depth  $d$  from a noisy vertical component of the gravity field  $g(s)$  measured at the surface. The forward model can be expressed by the first-kind Fredholm integral equation

$$g(s) = \int_a^b K(s, t) f(t) dt \quad (5.2)$$

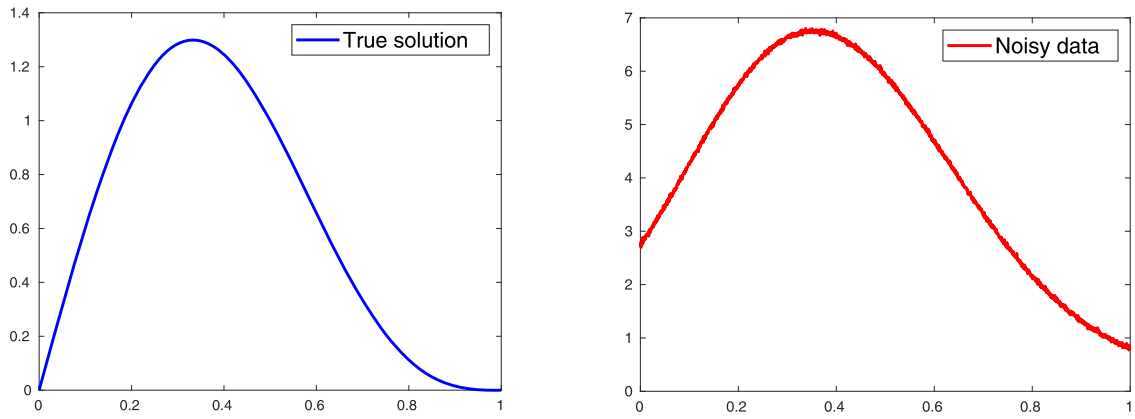


Fig. 1. Illustration of the true solution and noisy observed data for gravity.

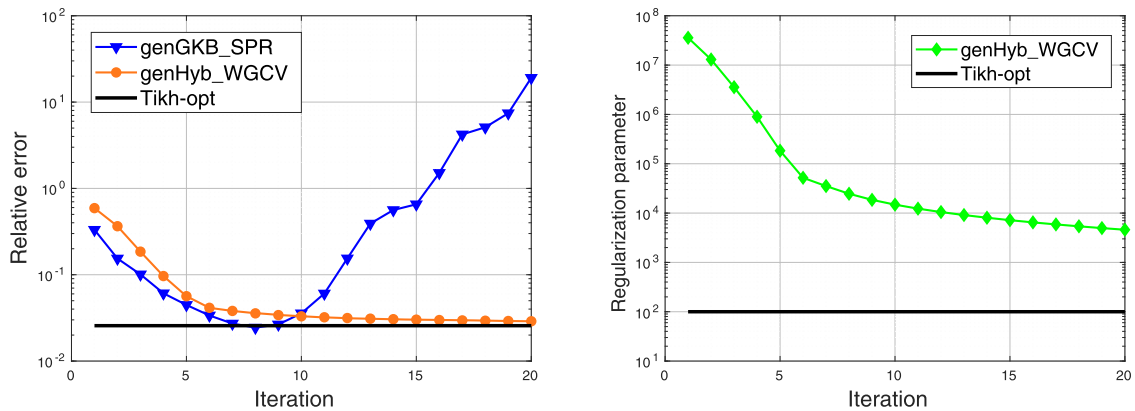


Fig. 2. Comparison of convergence behaviors between genGKB\_SPR and genHyb\_WGCV for gravity. Left: relative error curves of regularized solution with respect to iteration number; Right: convergence trend of regularization parameters of genHyb\_WGCV towards  $\lambda_{opt}$ .

with  $t \in [a, b] = [0, 1]$  and  $s \in [0, 1]$ , where the kernel  $K$  and true solution  $f$  are given by

$$K(s, t) = d(d^2 + (s - t)^2)^{-3/2}, \quad d = 0.25,$$

$$f(t) = \sin(\pi t) + 0.5 \sin(2\pi t).$$

This equation is discretized by the midpoint quadrature rule with  $m = n = 2000$  uniform points, leading to the matrix  $A \in \mathbb{R}^{2000 \times 2000}$ ,  $\mathbf{x}_{true} \in \mathbb{R}^{2000}$  and  $\mathbf{b}_{true} = A\mathbf{x}_{true}$ . We construct a discrete Gaussian white noise  $\epsilon \sim \mathcal{N}(\mathbf{0}, \sigma^2 I)$  such that the noisy level  $\mathbb{E}[\|\epsilon\|_2] / \|\mathbf{b}_{true}\|_2 \approx (\mathbb{E}[\|\epsilon\|_2^2] / \|\mathbf{b}_{true}\|_2^2)^{1/2} = \sqrt{m}\sigma / \|\mathbf{b}_{true}\|_2 = 5 \times 10^{-3}$ , and then let  $\mathbf{b} = \mathbf{b}_{true} + \epsilon$ . The true solution and noisy observed data are shown in Fig. 1.

To reconstruct  $\mathbf{x}$  from noisy  $\mathbf{b}$ , we assume a Gaussian prior  $\mathbf{x} \sim \mathcal{N}(\mathbf{0}, \lambda^{-1} N)$  with  $N$  coming from the Gaussian kernel  $\kappa_G$ , i.e. the  $ij$  element of  $N$  is

$$N_{ij} = \kappa_G(r_{ij}), \quad \kappa_G(r) := \exp(-r^2 / (2l^2)),$$

where  $r_{ij} = \|\mathbf{p}_i - \mathbf{p}_j\|_2$  and  $\{\mathbf{p}_i\}_{i=1}^n$  are discretized points of the domain of  $f$  (i.e.  $f(\mathbf{p}_i) = [\mathbf{x}_{true}]_i$ ). The parameter  $l$  is set as  $l = 0.1$ .

The convergence behaviors of genGKB\_SPR and genHyb\_WGCV are shown in Fig. 2 using the relative error curves. The final regularized solutions and the corresponding iteration number are shown in Table 1. In this experiment, all of the DP, LC and GCV methods under-estimate the semi-convergence point  $k_0$ , but the estimated early stopping iterations are only slightly smaller than  $k_0$ , and the reconstructed solutions approximate well to the true solution (Figs. 3 and 10).

For genHyb\_WGCV, the iterative solution tends to approximate the best Tikhonov regularization solution, but it takes much more iterations to reach a similar accuracy as the genGKB\_SPR solution at  $k_0$ . From the right subfigure of Fig. 2, we find that WGCV always determines a larger regularization parameter than  $\lambda_{opt}$  during the iteration, thereby computing an over-smoothed solution. Although there are currently no rigorous theoretical results, we can observe numerically that the convergence rate of the regularization parameter in genHyb\_WGCV is fast at first and then gradually slows down; this may be why the genHyb\_WGCV solution converges slowly to the best Tikhonov regularization solution. To compare the two methods, we can see from Table 1 that for genGKB\_SPR with DP, the solution at  $k = 6$  is only slightly less accurate than the genHyb\_WGCV solution at  $k = 20$ , while for genGKB\_SPR with LC or GCV the solution at  $k = 7$  has a slightly higher accuracy than the genHyb\_WGCV solution at  $k = 20$ .

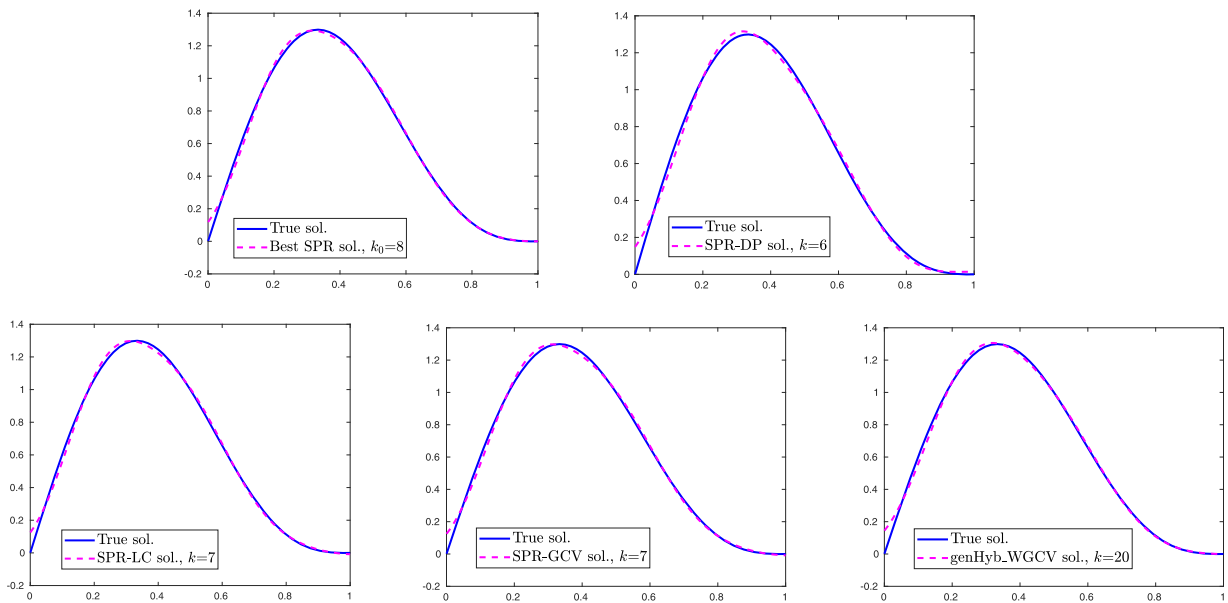


Fig. 3. Reconstructed solutions by genGKB\_SPR and genHyb\_WGCV for gravity.

Table 1

Relative error of the final regularized solution and corresponding early stopping iteration number (in parentheses) for gravity and shaw.

| Problem | SPR-best   | SPR-DP     | SPR-LC     | SPR-GCV    | genHyb_WGCV |
|---------|------------|------------|------------|------------|-------------|
| gravity | 0.0244 (8) | 0.0337 (6) | 0.0272 (7) | 0.0272 (7) | 0.0289 (20) |
| shaw    | 0.0487 (7) | 0.0613 (6) | 0.0983 (5) | 0.1706 (8) | 0.0761 (20) |

**Experiment 2.** The second small-scale problem is shaw that models a one-dimensional image restoration using the Fredholm integral equation (5.2), where the kernel  $K$  and solution  $f$  are given by

$$K(s, t) = (\cos s + \cos t)^2 \left( \frac{\sin u}{u} \right)^2, \quad u = \pi(\sin s + \sin t),$$

$$f(t) = 2 \exp(-6(t - 0.8)^2) + \exp(-2(t + 0.5)^2).$$

The discretization approach is the same as gravity and  $m = n = 2000$ . We constructed a non-white Gaussian noise  $\epsilon \sim \mathcal{N}(\mathbf{0}, \mathbf{M})$  as follows such that  $\mathbf{M}$  is a diagonal matrix. First formulate a vector  $\mathbf{d} = (d_1, \dots, d_m)$  with elements of uniform random integers between  $[1, 5]$ . Then let  $\mathbf{M} = \text{diag}(\gamma \mathbf{d})$ , where  $\gamma$  satisfies that the noise level  $\mathbb{E}[\|\epsilon\|_2] / \|\mathbf{b}_{\text{true}}\|_2 \approx (\mathbb{E}[\|\epsilon\|_2^2] / \|\mathbf{b}_{\text{true}}\|_2^2)^{1/2} = (\gamma \sum_{i=1}^m d_i)^{1/2} / \|\mathbf{b}_{\text{true}}\|_2 = 10^{-2}$ . The true solution and noisy observed data are shown in Fig. 4. The covariance matrix  $\mathbf{N}$  of the Gaussian prior of  $\mathbf{x}$  comes from the exponential kernel

$$\kappa_{\text{exp}}(r) := \exp(-(r/l)^\nu),$$

and the parameters  $l$  and  $\nu$  are set as  $l = 0.1$  and  $\nu = 1$ .

The convergence behaviors of genGKB\_SPR and genHyb\_WGCV for shaw are shown in Fig. 5, while the reconstructed solutions are drawn in Fig. 6. We can observe similar phenomena as the first experiment: the genGKB\_SPR solution exhibits the semi-convergence property, while the genHyb\_WGCV gradually converges to the best Tikhonov regularization solution and it takes much more iterations to reach a similar accuracy as the genGKB\_SPR solution at  $k_0$ . From Table 1 we find that DP and LC slightly under-estimate  $k_0$  while GCV over-estimate  $k_0$ . In this experiment, the relative error for genGKB\_SPR increases rapidly after the semi-convergence point, resulting in the GCV solution at  $k = 8$  with a relatively poor accuracy. However, the DP or LC solution of genGKB\_SPR at  $k = 6$  or  $k = 5$  has a satisfied accuracy, which is comparable to the genHyb\_WGCV solution at  $k = 20$ .

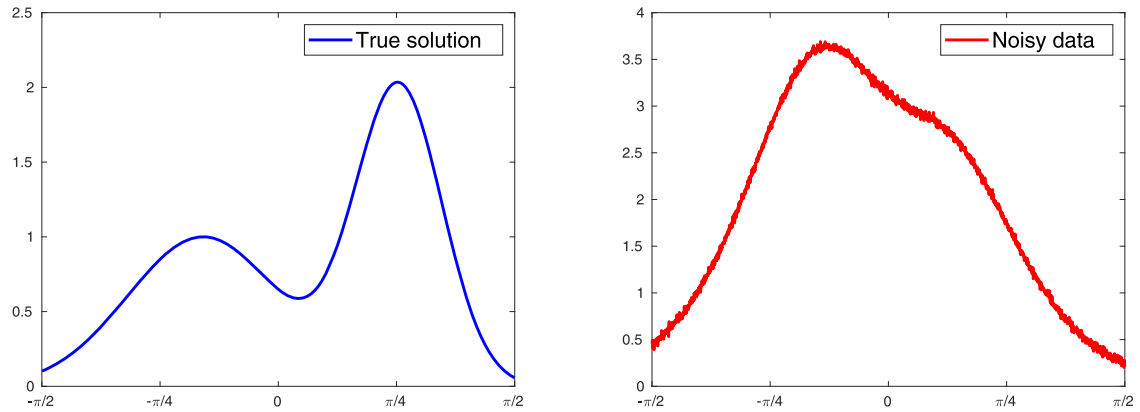


Fig. 4. Illustration of the true solution and noisy observed data for shaw.

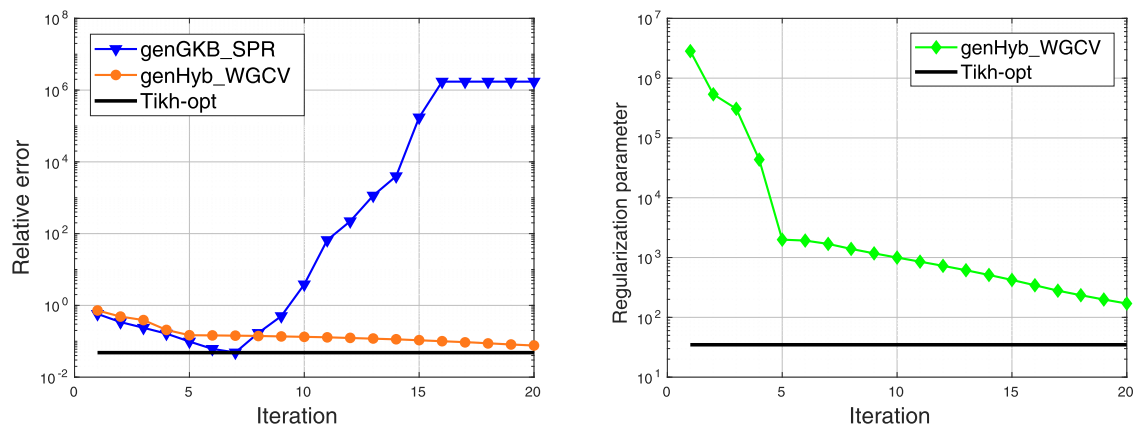


Fig. 5. Comparison of convergence behaviors between genGKB\_SPR and genHyb\_WGCV for shaw. Left: relative error curves of regularized solution with respect to iteration number; Right: convergence trend of regularization parameters of genHyb\_WGCV towards  $\lambda_{opt}$ .

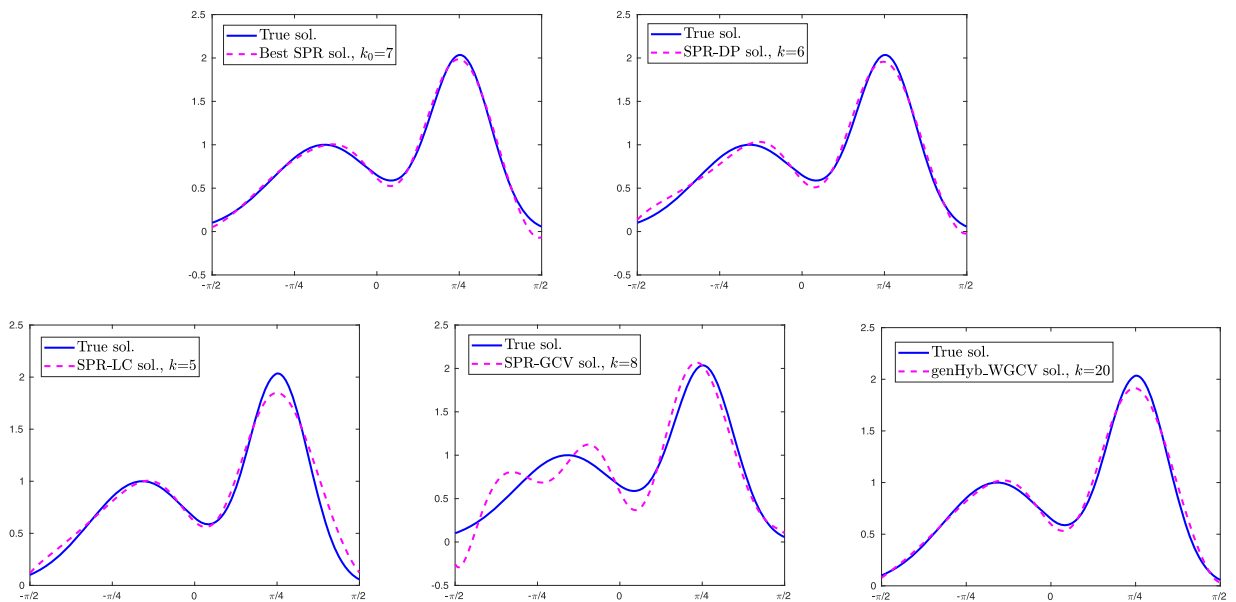


Fig. 6. Reconstructed solutions by genGKB\_SPR and genHyb\_WGCV for shaw.

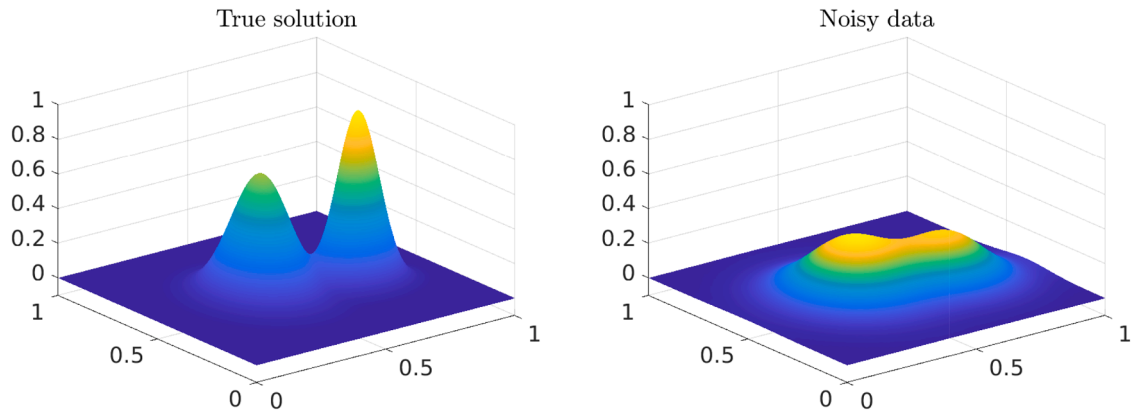


Fig. 7. Illustration of the true solution and noisy observed data for PRdiffusion.

### 5.2. Two-dimensional inverse diffusion

In this experiment, we consider the two-dimensional (2D) inverse diffusion problem PRdiffusion [38]. The forward problem is a 2D diffusion equation in the spatial domain  $\Omega = [0, 1]^2$  with homogenous Neumann boundary condition and an initial function  $u_0$ :

$$\begin{cases} \frac{\partial u}{\partial t} = \Delta u, & (x, y) \in \Omega, \quad t > 0, \\ \frac{\partial u}{\partial \mathbf{n}} = 0, & (x, y) \in \partial\Omega, \quad t > 0, \\ u(0, x, y) = u_0(x, y), & (x, y) \in \Omega, \end{cases}$$

where  $\mathbf{n}$  denotes the exterior normal to the boundary  $\partial\Omega$ . The inverse problem is to reconstruct  $u_0$  from the diffusion solution  $u_T = u(T, x, y)$  at time  $t = T$ . In the experiment, we set  $T = 0.005$  and  $u_0$  be the following smooth function:

$$u_0(x, y) = 0.7 \exp\left(-\left(\frac{x-0.4}{0.12}\right)^2 - \left(\frac{y-0.5}{0.15}\right)^2\right) + \exp\left(-\left(\frac{x-0.7}{0.1}\right)^2 - \left(\frac{y-0.8}{0.08}\right)^2\right).$$

The above forward PDE is discretized on a uniform finite element mesh with  $2(n_1 - 1)^2$  triangular elements, thereby  $u$  is represented by the  $n = n_1^2$  values at the corners of the elements, and  $[0, T]$  is divided into 100 uniform time steps. The forward computation is the numerical solution of the above PDE implemented by the Crank-Nicolson-Galerkin finite element method, thereby  $\mathbf{A}$  is a functional handle. By choosing  $n_1 = 128$ , we have  $\mathbf{A} \in \mathbb{R}^{128^2 \times 128^2}$ , which maps  $\mathbf{x}_{\text{true}}$  (the discretized  $u_0$ ) to  $\mathbf{b}_{\text{true}}$  (the discretized  $u_T$ ), and we add a Gaussian white noise  $\epsilon$  with noise level  $10^{-3}$  to  $\mathbf{b}_{\text{true}}$  to get  $\mathbf{b}$ . The true solution and noisy observed data are shown in Fig. 7.

In order to reconstruct the smooth initial function  $u_0$ , we assume a Gaussian prior of  $\mathbf{x}$  with the covariance matrix  $\mathbf{N}$  coming from the Matérn kernel

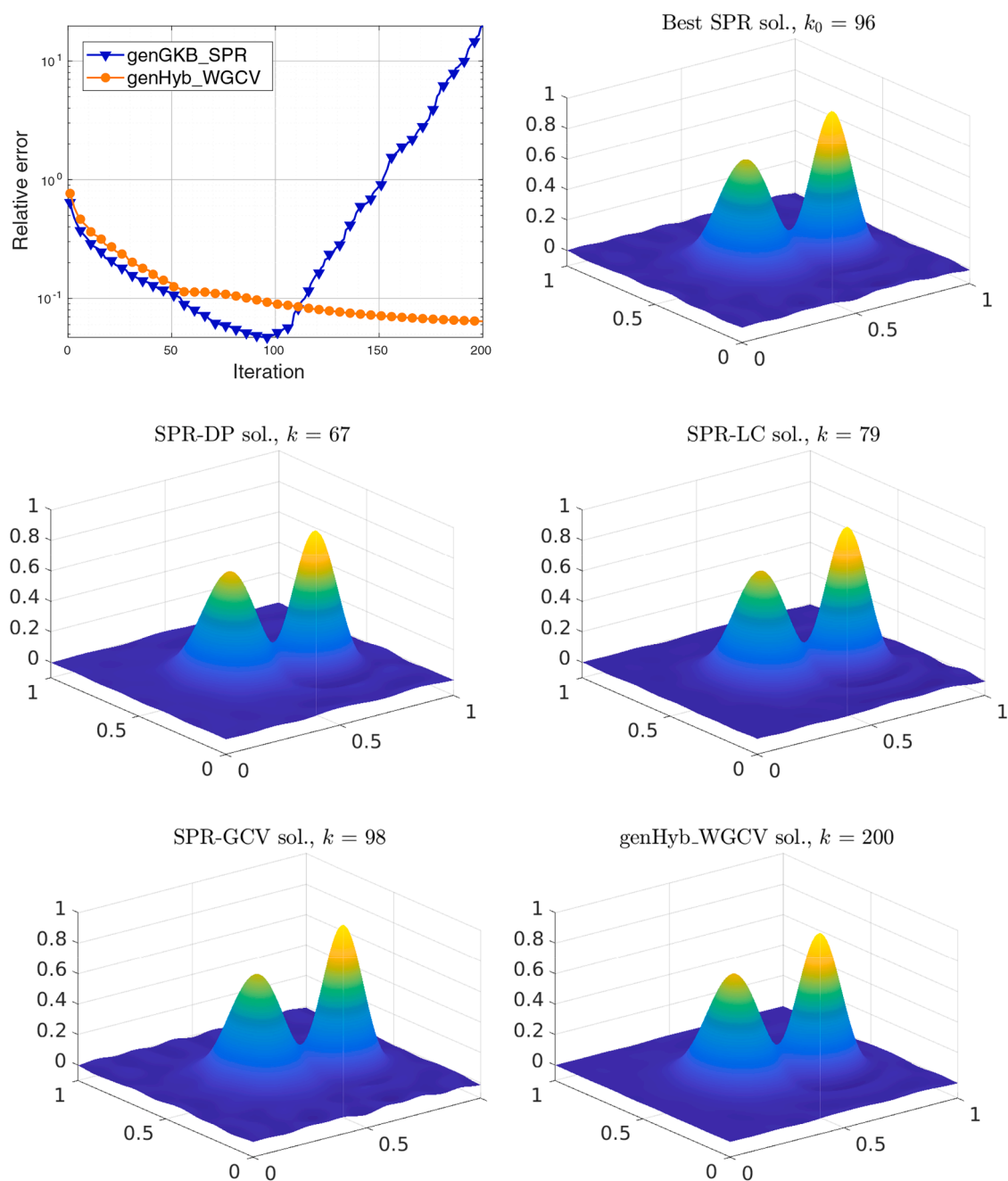
$$\kappa_M(r) := \frac{2^{1-\nu}}{\Gamma(\nu)} \left( \frac{\sqrt{2\nu}r}{l} \right)^\nu K_\nu \left( \frac{\sqrt{2\nu}r}{l} \right),$$

where  $\Gamma$  is the gamma function,  $K_\nu$  is the modified Bessel function of the second kind, and  $l$  and  $\nu$  are two positive parameters of the covariance. The parameter  $\nu$  controls the smoothness of the resulting function; see Williams and Rasmussen [39, §4.2]. The smaller the value of  $\nu$ , the less smooth the approximated function becomes. We set  $l = 0.05$  and  $\nu = 5/2$  (for twice differentiable functions), respectively.

From Fig. 8 we find genGKB\_SPR exhibits the semi-convergence behavior and the best regularized solution is obtained at  $k_0 = 96$ . The estimates of  $k_0$  by DP, LC and GCV and the relative errors of corresponding solutions are shown in Table 2. In comparison, the genHyb\_WGCV solution convergences very slowly; its solution only reaches an accuracy slightly poorer than the best genGKB\_SPR solution even for  $k = 200$ , much larger than  $k_0$ . As shown in Table 2, the DP, LC, and GCV solutions of genGKB\_SPR also have higher accuracy than the genGKB\_SPR solution at  $k = 200$ . Considering this, for large-scale problems, it will take much more time for genHyb\_WGCV than genGKB\_SPR to obtain a satisfied regularized solution. We find that the reconstructed solutions by genGKB\_SPR have good smoothness and a higher similarity to  $u_0$ . This is due to the elaborately constructed solution subspaces by gen-GKB that incorporate prior information of  $\mathbf{x}$  encoded by the Matérn kernel to enforce the second order differentiability of a desired  $\mathbf{x}$ .

### 5.3. Two-dimensional spherical radon transform tomography

In this experiment, we test our method by a 2D spherical Radon transform (SRT) tomography. The SRT integrates a function over a set of spheres (or circles in the 2D case) to get a noisy observation, which plays an important role in some newly developing types of tomography such as photo-acoustic imaging [40]. Here we consider the 2D SRT problem where the forward operator is



**Fig. 8.** Convergence behaviors and reconstructed solutions by genGKB\_SPR and genHyb\_WGCV for PRdiffusion.

**Table 2**

Relative error of the final regularized solution and corresponding early stopping iteration number (in parentheses) for PRdiffusion and PRspherical.

| Problem     | SPR-best    | SPR-DP      | SPR-LC      | SPR-GCV     | genHyb_WGCV  |
|-------------|-------------|-------------|-------------|-------------|--------------|
| PRdiffusion | 0.0470 (96) | 0.0704 (67) | 0.0566 (79) | 0.0483 (98) | 0.0641 (200) |
| shaw        | 0.0026 (25) | 0.0046 (17) | 0.0033 (20) | 0.0156 (67) | 0.0106 (100) |

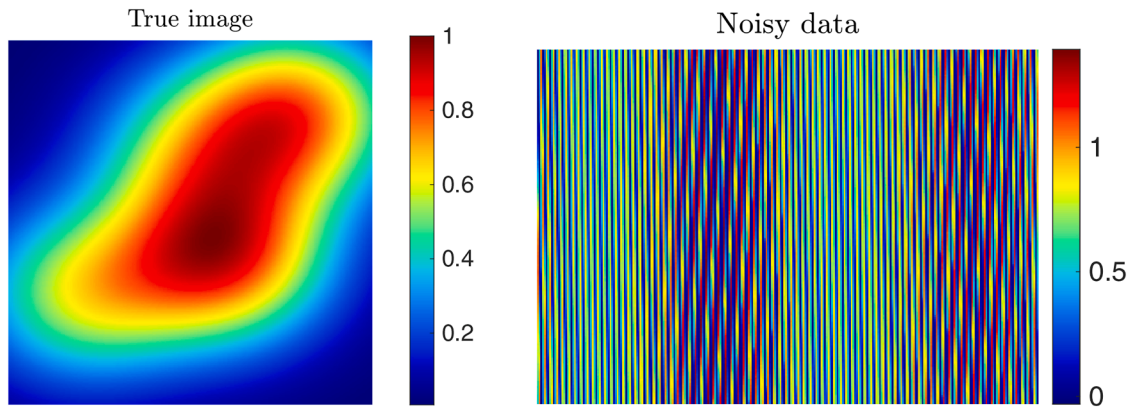


Fig. 9. Illustration of the true image and noisy observed data for PRspherical.

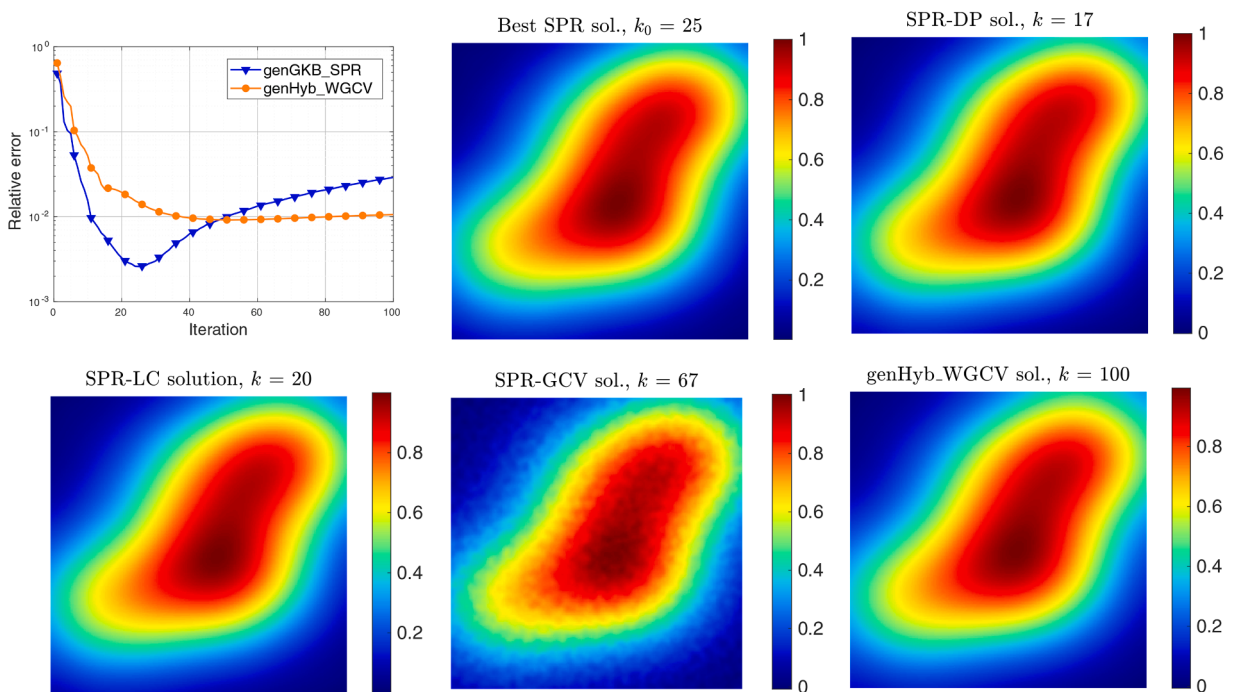


Fig. 10. Convergence behaviors and reconstructed solutions by genGKB\_SPR and genHyb\_WGCV for PRspherical.

generated by PRspherical with default settings [38]; for detailed descriptions of PRspherical see Hansen and Jørgensen [41, §3.2]. The true solution is a  $256 \times 256$  pixel 2D smooth image consisting of a superposition of four Gaussian functions. Following the same approach as the experiment for shaw, we add a Gaussian non-white noise  $\epsilon$  with noise level  $10^{-2}$  to  $b_{\text{true}}$  to get  $b$ . In the experiment  $A \in \mathbb{R}^{92672 \times 65536}$ . The true image and noisy observed data are shown in Fig. 7. In order to reconstruct a smooth image, we assume a Gaussian prior of  $x$  with Matérn kernel, where the two parameters are set as  $l = 500$  and  $\nu = 1/2$ , respectively.

The convergence behaviors of the two methods and corresponding reconstructed solutions are shown in Fig. 9. For this large-scale inverse problems, we can find that the genGKB\_SPR method only takes 25 steps to reach the semi-convergence point, while even after 100 steps, genHyb\_WGCV only obtains a solution with significantly lower accuracy than the best genGKB\_SPR solution. If we carefully observe the convergence curve, we can notice that the relative error of the genHyb\_WGCV solution slightly increases as  $k$  grows from 60 to 100. This potential instability of convergence for the hybrid methods has already been pointed out in Chung et al. [22] and Renaut et al. [23], which is caused by the instability of determining regularized parameter for the projected problem (5.1) regardless of the parameter choosing method used. Although the estimates of  $k_0$  for genGKB\_SPR by DP and LC are slightly smaller than the true value, resulting in two over-smoothed solutions, we can still observe that the reconstructed images have satisfactory quality. The relative errors of them are both smaller than the genHyb\_WGCV solution at  $k = 100$  (also smaller than the best genHyb\_WGCV

solution at around  $k = 60$ ). However, in this experiment, the GCV method suffers severely from overestimating the true  $k_0$ , and the corresponding solution deviates significantly from the best genGKB\_SPR, resulting in a very poor accuracy.

Overall, the aforementioned experiments have confirmed that the genGKB\_SPR algorithm combined with a proper early stopping rule provides a viable alternative for directly solving the Bayesian linear inverse problems (1.2). The numerical tests have shown that genGKB\_SPR with an early stopping rule is more computationally effective and robust for reconstructing solutions compared to the hybrid method genHyb\_WGCV, especially for large-scale problems.

## 6. Conclusion

In this paper, we have proposed a new iterative regularization algorithm for solving Bayesian linear inverse problems. Within the subspace projection regularization framework, we first introduced an iterative process that can generate a series of valid solution subspaces by treating the forward model matrix as a linear operator between the two Hilbert spaces  $(\mathbb{R}^n, \langle \cdot, \cdot \rangle_{N^{-1}})$  and  $(\mathbb{R}^m, \langle \cdot, \cdot \rangle_{M^{-1}})$ . The original problem is projected onto these subspaces becoming a series of low dimensional linear least squares problems. We have developed an efficient procedure to update the solutions of these projected problems as approximate iterative regularization solutions, as well as update the residual norm and solution norm of them. Early stopping rules based on DP, LC and GCV have been designed, with which the iterative algorithm can get a regularized solution to the original problem with a satisfied accuracy. Several theoretical results have been established, revealing the good regularization properties of the algorithm. Numerical experiments using both small-scale and large-scale inverse problems have been conducted to demonstrate the robustness and efficiency of the algorithm. Since the most computationally intensive operations in the proposed algorithm only involve matrix-vector products, it is highly efficient for large-scale Bayesian inverse problems.

## CRedit authorship contribution statement

**Haibo Li:** Writing – review & editing, Writing – original draft, Methodology, Formal analysis, Conceptualization.

## Declaration of competing interest

None.

## References

- [1] J. Kaipio, E. Somersalo, *Statistical and Computational Inverse Problems*, Springer, 2006.
- [2] P.C. Hansen, J.G. Nagy, D.P. O’Leary, *Deblurring Images: Matrices, Spectra and Filtering*, SIAM, Philadelphia, 2006.
- [3] T. Buzug, *Computed Tomography*, Springer, 2008. <https://doi.org/10.1007/978-3-540-39408-2>.
- [4] K. Law, A. Stuart, K. Zygalakis, *Data Assimilation: A Mathematical Introduction*, Springer, 2015. <https://doi.org/10.1007/978-3-319-20325-6>.
- [5] M. Richter, *Inverse Problems: Basics, Theory and Applications in Geophysics*, Springer, 2016. [https://doi.org/10.1007/978-3-319-48384-9\\_3](https://doi.org/10.1007/978-3-319-48384-9_3).
- [6] J. Hadamard, *Lectures on Cauchy’s Problem In Linear Partial Differential Equations*, 15, Yale university press, 1923.
- [7] A.M. Stuart, Inverse problems: a Bayesian perspective, *Acta Numer.* 19 (2010) 451–559.
- [8] A.N. Tikhonov, V.Y. Arsenin, *Solutions of Ill-Posed Problems*, Winston & Sons (distributed by Halsted Press, a division of John Wiley & Sons), Washington, DC, 1977.
- [9] P.C. Hansen, Analysis of discrete ill-posed problems by means of the L-curve, *SIAM Rev.* 34 (4) (1992) 561–580.
- [10] G.H. Golub, H.G. Wahba, Generalized cross-validation as a method for choosing a good ridge parameter, *Technometrics* 21 (2) (1979) 215–223.
- [11] V.A. Morozov, Regularization of incorrectly posed problems and the choice of regularization parameter, *USSR Comput. Math. Math. Phys.* 6 (1) (1966) 242–251.
- [12] J. Liesen, Z. Strakos, Krylov subspace methods: principles and analysis, *Numer. Math. Sci.* 52 (1) (2015) 151–158.
- [13] D. Calvetti, F. Pitolli, E. Somersalo, B. Vantaggi, Bayes meets Krylov: statistically inspired preconditioners for CGLS, *SIAM Rev.* 60 (2) (2018) 429–461.
- [14] D. Calvetti, E. Somersalo, Priorconditioners for linear systems, *Invers. Probl.* 21 (4) (2005) 1397.
- [15] D. Calvetti, F. Pitolli, J. Prezioso, E. Somersalo, B. Vantaggi, Priorconditioned CGLS-based quasi-MAP estimate, statistical stopping rule, and ranking of priors, *SIAM J. Sci. Comput.* 39 (5) (2017) S477–S500.
- [16] M.G. Genton, Classes of kernels for machine learning: a statistics perspective, *J. Mach. Learn. Res.* 2 (Dec) (2001) 299–312.
- [17] L. Roininen, M.S. Lehtinen, S. Lasanen, M. Orispää, M. Markkanen, Correlation priors, *Invers. Probl. Imag.* 5 (1) (2011) 167–184.
- [18] H. Antil, A.K. Saibaba, Efficient algorithms for Bayesian inverse problems with Whittle–Matérn priors, *SIAM J. Sci. Comput.* (2023) S176–S198.
- [19] J. Chung, A.K. Saibaba, Generalized hybrid iterative methods for large-scale Bayesian inverse problems, *SIAM J. Sci. Comput.* 39 (5) (2017) S24–S46.
- [20] M. Arioli, Generalized Golub–Kahan bidiagonalization and stopping criteria, *SIAM J. Matrix Anal. Appl.* 34 (2) (2013) 571–592. <https://doi.org/10.1137/120866543>
- [21] M.E. Kilmer, D.P. O’Leary, Choosing regularization parameters in iterative methods for ill-posed problems, *SIAM J. Matrix Anal. Appl.* 22 (4) (2001) 1204–1221.
- [22] J. Chung, J.G. Nagy, D.P. O’Leary, A weighted-GCV method for Lanczos-hybrid regularization, *Electr. Trans. Numer. Anal.* 28 (29) (2008) 149–167.
- [23] R.A. Renaut, S. Vatankehah, V.E. Ardesta, Hybrid and iteratively reweighted regularization by unbiased predictive risk and weighted GCV for projected systems, *SIAM J. Sci. Comput.* 39 (2) (2017) B221–B243.
- [24] H.W. Engl, M. Hanke, A. Neubauer, *Regularization of Inverse Problems*, Kluwer Academic Publishers, 2000.
- [25] P.C. Hansen, Regularization, GSVD and truncated GSVD, *BIT Numer. Math.* 29 (3) (1989) 491–504.
- [26] G.H. Golub, C.F. Van Loan, *Matrix Computations*, The Johns Hopkins University Press, Baltimore, 4th edition, 2013.
- [27] P.C. Hansen, The discrete picard condition for discrete ill-posed problems, *BIT Numer. Math.* 30 (4) (1990) 658–672.
- [28] N.A. Caruso, P. Novati, Convergence analysis of LSQR for compact operator equations, *Linear Algebr. Appl.* 583 (2019) 146–164.
- [29] C.C. Paige, M.A. Saunders, LSQR: an algorithm for sparse linear equations and sparse least squares, *ACM Trans. Math. Softw.* 8 (1982) 43–71.
- [30] P.C. Hansen, Rank-Deficient and Discrete Ill-Posed Problems: Numerical Aspects of Linear Inversion, SIAM, Philadelphia, 1998.
- [31] G.W. Stewart, *Matrix Algorithms, Volume II: Eigensystems*, SIAM, 2001.
- [32] Å. Björck, *Numerical Methods in Matrix Computations*, 59, Springer, 2015.
- [33] A.V. Knyazev, M.E. Argentati, Principal angles between subspaces in an a-based scalar product: algorithms and perturbation estimates, *SIAM J. Sci. Comput.* 23 (6) (2002) 2008–2040. <https://doi.org/10.1137/S1064827500377332>
- [34] P.C. Hansen, *Discrete Inverse Problems: Insight and Algorithms*, SIAM, Philadelphia, 2010.

- [35] Z. Bai, J. Demmel, J. Dongarra, A. Ruhe, H. van der Vorst, *Templates for the Solution of Algebraic Eigenvalue Problems: A Practical Guide*, SIAM, 2000.
- [36] A. Van der Sluis, H.A. van der Vorst, The rate of convergence of conjugate gradients, *Numer. Math.* 48 (1986) 543–560.
- [37] P.C. Hansen, Regularization tools version 4.0 for Matlab 7.3, *Numer. Algor.* 46 (2) (2007) 189–194.
- [38] S. Gazzola, P.C. Hansen, J.G. Nagy, IR Tools: a MATLAB package of iterative regularization methods and large-scale test problems, *Numer. Algor.* 81 (3) (2019) 773–811.
- [39] C.K.I. Williams, C.E. Rasmussen, *Gaussian Processes for Machine Learning*, MIT press Cambridge, MA, 2006.
- [40] J. Klukowska, R. Davidi, G.T. Herman, SNARK09—a software package for reconstruction of 2D images from 1D projections, *Comput. Method. Program. Biomed.* 110 (3) (2013) 424–440.
- [41] P.C. Hansen, J.S. Jørgensen, AIR Tools II: algebraic iterative reconstruction methods, improved implementation, *Numer. Algor.* 79 (1) (2018) 107–137.